



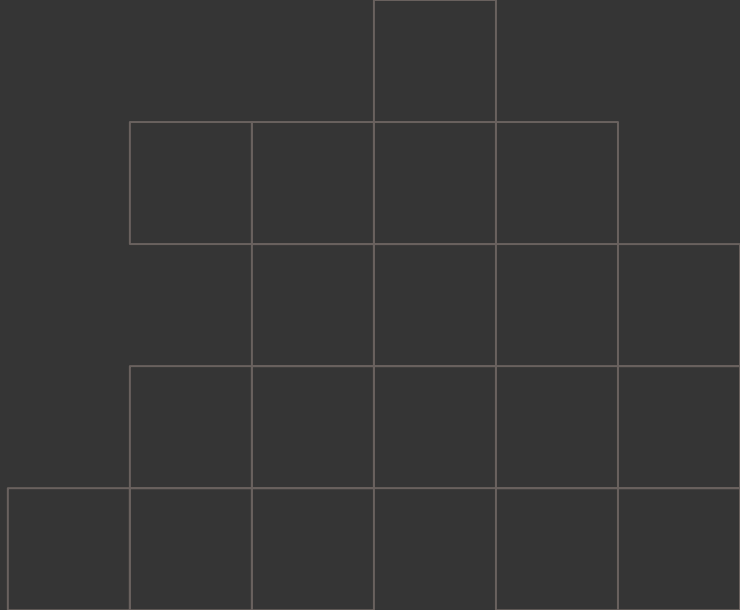
Lessons learned from maintaining OSS projects

About this talk

For 100 developers using OSS

10 contributors

1 maintainer

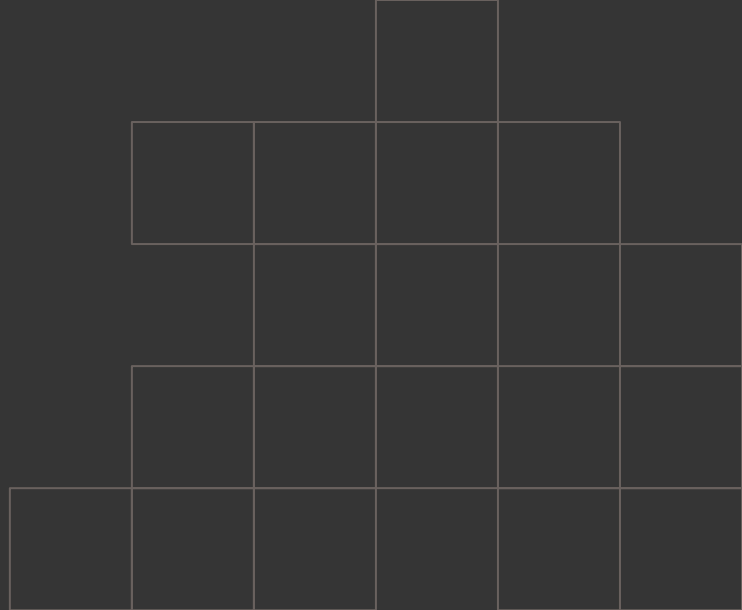


About this talk

For 100 developers using OSS

10 contributors

1 maintainer <- This talk



\$ cat summary.txt



Developer experience



Contributor experience



Marketing



\$\$\$



Security

whoami

My name is Samuel.

> Ex-CTO @ Screeb

> GenAI Nantes & Shift organizer

> Full-time open-sourcer
github.com/samber



My open-source projects



Samuel Berthe

samber · he/him

Software Engineer | Go Open-sourcer |
GenAI community leader

Pinned

[Customize your pins](#)

 **lo** Public

🔥 A Lodash-style Go library based on Go 1.18+ Generics (map, filter, contains, find...)

Go ☆ 20.8k 🍴 926

 **do** Public

⚙️ A dependency injection toolkit based on Go 1.18+ Generics.

Go ☆ 2.6k 🍴 103

 **mo** Public

🔥 Monads and popular FP abstractions, powered by Go 1.18+ Generics (Option, Result, Either...)

Go ☆ 3.3k 🍴 109

 **ro** Public

🔥 Streams & Reactive Programming paradigm for Go: declarative and composable API for event-driven applications

Go ☆ 564 🍴 17

 **awesome-prometheus-alerts** Public

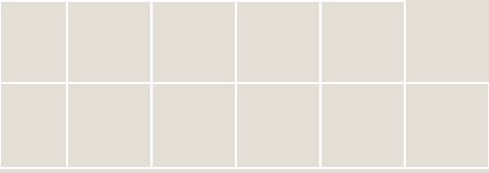
🔥 Collection of Prometheus alerting rules

HTML ☆ 7.7k 🍴 1.1k

 **oops** Public

🔥 Error handling library with context, assertion, stack trace and source fragments

Go ☆ 839 🍴 34

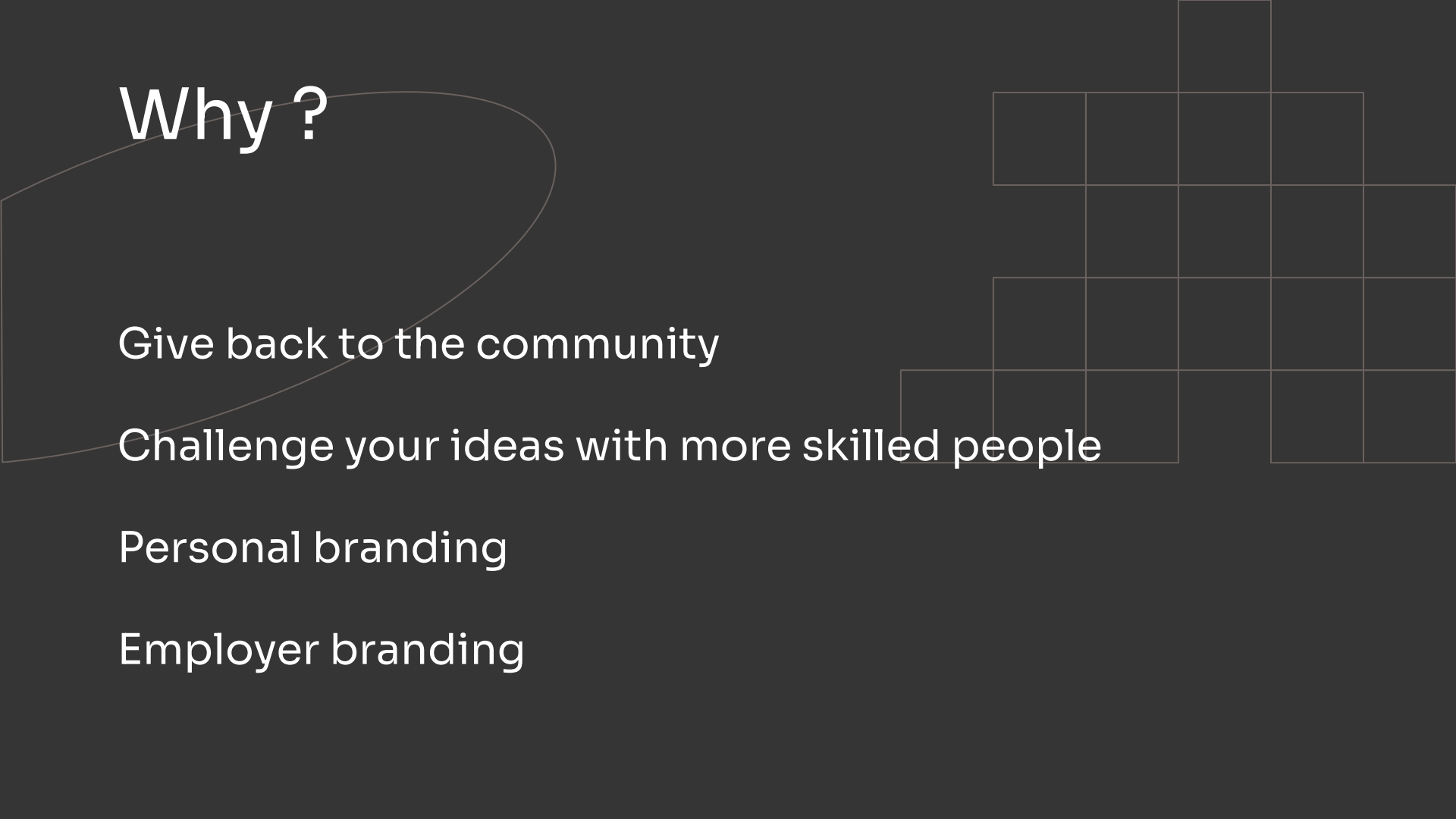


1- Open-source software

What and why ?



Why ?



Give back to the community

Challenge your ideas with more skilled people

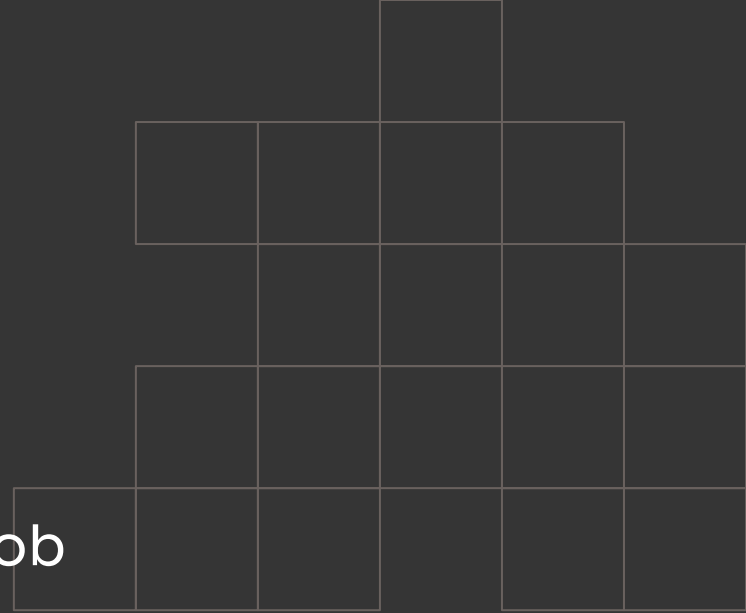
Personal branding

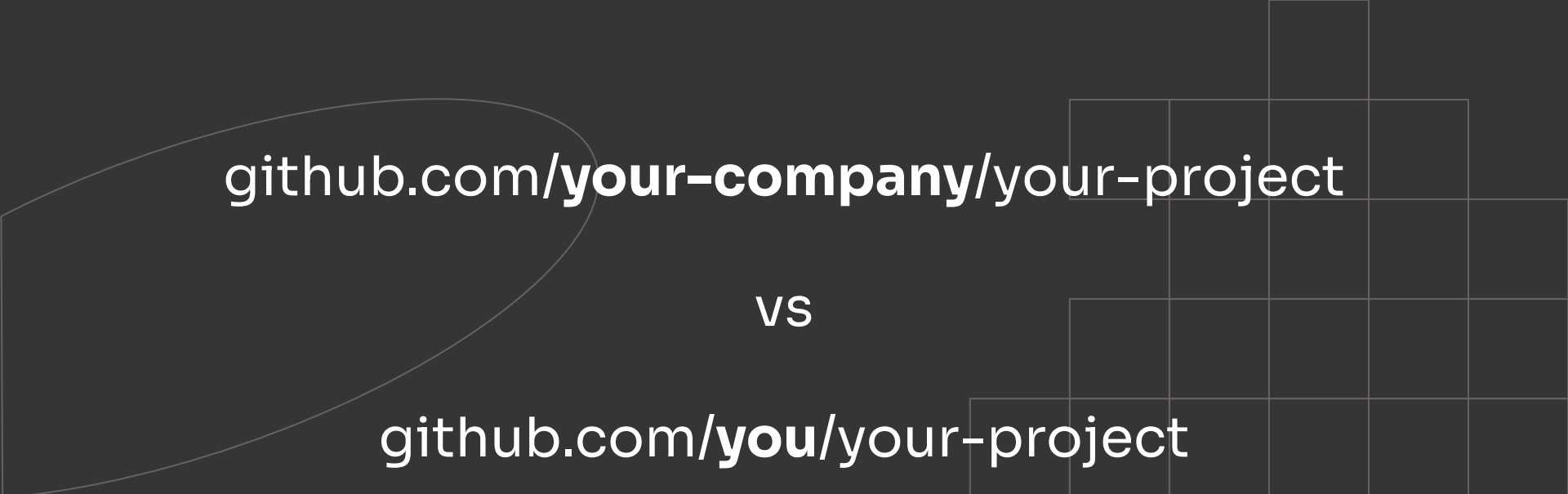
Employer branding

How to start ?

Publish tools you built in your daily job

Not related to you company IP





github.com/**your-company**/your-project

VS

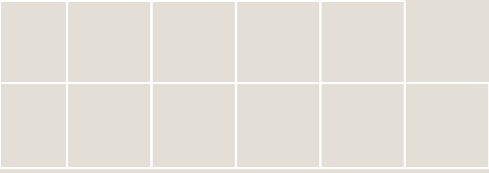
github.com/**you**/your-project

VS

github.com/**your-community**/your-project

License

License	Type	Main Obligations	Commercial Use	Copyleft?
MIT	Very permissive	Keep copyright and license notice	✅ Yes	❌ No
BSD (2- or 3-clause)	Very permissive	Keep copyright; sometimes forbid using author's name for promotion	✅ Yes	❌ No
Apache 2.0	Permissive with patent protection	Attribution + note modifications + patent clause	✅ Yes	❌ No
AGPL v3	Strong copyleft (includes network use)	Share complete source code, including for web apps	⚠️ Hard (incompatible with proprietary software)	✅ Yes, strong



2- Take care of developer experience





Invest time in your README.

Oops - Structured Error Handling for Go

Transform your Go error handling from "oops!" to "aha!" moments

tag `v1.19.3` Go `>= 1.21` `go` reference  Tests `passing` go report `A+` coverage `79%` contributors `10` license `MIT`

Oops is a comprehensive Go error handling library that provides structured error management with rich contextual information. It's designed as a drop-in replacement for Go's built-in `error`, adding powerful features like stack traces, source code fragments, structured attributes, and developer-friendly debugging hints.

Key Features:

-  **Drop-in Replacement:** Seamlessly replaces standard Go error handling
-  **Rich Context:** Add structured attributes, user info
-  **Debug-Friendly:** Out-of-the-box stacktraces and source code fragments
-  **Error Chaining:** Wrap and compose errors with additional context
-  **Panic Recovery:** Built-in panic handling with error conversion
-  **Assertions:** One-line assertion helpers for validation
-  **Performance:** Zero dependencies, lightweight and fast
-  **Logger Integration:** Works with all major Go logging libraries
-  **Separation of Concerns:** Error handling and logging are separate jobs
-  **Easy Integration:** No large refactoring required

Warning

Important: This is NOT a logging library. `oops` should complements your existing logging toolchain (zap, zerolog, logrus, slog, go-sentry...).

 Start hacking `oops` with this [playground](#).



The Go Playground

Go 1.25

Run

Format

Share

Hello, World!

Press Esc to move out of the editor.

```
1 package main
2
3 import (
4     "time"
5
6     "github.com/samber/oops"
7     oopslogrus "github.com/samber/oops/loggers/logrus"
8     "github.com/sirupsen/logrus"
9 )
10
11 func main() {
12     logrus.SetFormatter(oopslogrus.NewOopsFormatter(&logrus.JSONFormatter{
13         PrettyPrint: true,
14     }))
15
16     err := oops.
17         Code("iam_authz_missing_permission").
18         In("authz").
19         Time(time.Now()).
20         With("user_id", 1234).
21         With("permission", "post.create").
22         Hint("Runbook: https://doc.acme.org/doc/abcd.md").
23         User("user-123", "firstname", "john", "lastname", "doe").
24         Errorf("permission denied")
25
26     logrus.WithError(err).Error(err)
27 }
```

```
{
  "code": "iam_authz_missing_permission",
  "context": {
    "permission": "post.create",
    "user_id": 1234
  }
}
```

Context rich error handling

In a few minutes, your logs will look like this:

```
{
  "time": "2023-05-02 05:26:48.570837Z",
  "level": "ERROR",
  "msg": "something failed: permission denied",
  "error": {
    "code": "iam_missing_permission",
    "context": {
      "hello": "world",
      "permission": "post.create",
      "user_id": 1234
    },
    "domain": "authz",
    "tags": ["iam", "authz"],
    "error": "something failed: permission denied",
    "hint": "Runbook: https://doc.acme.org/doc/abcd.md",
    "stacktrace":
      "Oops: permission denied
      --- at github.com/samber/oops/loggers/slog/example.go:20 (d)
      --- at github.com/samber/oops/loggers/slog/example.go:24 (c)
      Thrown: something failed
      --- at github.com/samber/oops/loggers/slog/example.go:32 (b)
      --- at github.com/samber/oops/loggers/slog/example.go:36 (a)
      --- at github.com/samber/oops/loggers/slog/example.go:42 (main)",
    "time": "2023-05-02 05:26:48.570837Z",
    "trace": "e76031ee-a0c4-4a80-88cb-17086fdd19c0",
```

Install

```
go get github.com/samber/oops
```



This library is v1 and follows SemVer strictly.

No breaking changes will be made to APIs before v2.0.0.

This library has no dependencies outside the Go standard library.

Quick start

This library provides a simple `error` builder for composing structured errors, with contextual attributes and stack trace.

Since `oops.OopsError` implements the `error` interface, you will be able to compose and wrap native errors with `oops.OopsError`.

```
import "github.com/samber/oops"

func main() {
    // Simple error with context
    err := oops.
        In("user-service").
        Tags("database", "postgres").
        Code("network_failure").
        User("user-123", "email", "foo@bar.com").
        With("path", "/hello/world").
        Errorf("failed to fetch user: %s", "connection timeout")

    // Error wrapping
    if err != nil {
        return oops.
            Trace("req-123").
            With("product_id", "456").
            Wrapf(err, "user operation failed")
    }
}
```



Error builder

Method	Description
<code>.New(message string) error</code>	Formats returns <code>oops.OopsError</code> object that satisfies <code>error</code>
<code>.Errorf(format string, args ...any) error</code>	Formats an error and returns <code>oops.OopsError</code> object that satisfies <code>error</code>
<code>.Wrap(err error) error</code>	Wraps an error into an <code>oops.OopsError</code> object that satisfies <code>error</code>
<code>.Wrapf(err error, format string, args ...any) error</code>	Wraps an error into an <code>oops.OopsError</code> object that satisfies <code>error</code> and formats an error message
<code>.Recover(cb func()) error</code>	Handle panic and returns <code>oops.OopsError</code> object that satisfies <code>error</code> .
<code>.Recoverf(cb func(), format string, args ...any) error</code>	Handle panic and returns <code>oops.OopsError</code> object that satisfies <code>error</code> and formats an error message.
<code>.Assert(condition bool) OopsErrorBuilder</code>	Panics if condition is false. Assertions can be chained.
<code>.Assertf(condition bool, format string, args ...any) OopsErrorBuilder</code>	Panics if condition is false and formats an error message. Assertions can be chained.
<code>.Join(err1 error, err2 error, ...) error</code>	Join returns an error that wraps the given errors.

Examples

```
// with error wrapping
err0 := oops.
    In("repository").
    Tags("database", "sql").
    Wrapf(sql.Exec(query), "could not fetch user") // Wrapf returns nil when sql.Exec() is nil

// with panic recovery
err1 := oops.
    In("repository").
    Tags("database", "sql").
    Recoverf(func() {
```



Contributing

- Ping me on Twitter [@samuelberthe](#) (DMs, mentions, whatever :))
- Fork the [project](#)
- Fix [open issues](#) or request new features

Don't hesitate ;)

```
# Install some dev dependencies
make tools

# Run tests
make test
# or
make watch-test
```



Contributors



Show your support

Give a  if this project helped you!

SPONSORS

8



License

Copyright © 2023 [Samuel Berthe](#).

This project is [MIT](#) licensed.

Invest time in your README.

Notes:

- Don't list everything -> be useful
- Ask your ~~grandmother~~ intern to read the doc
- Ask ChatGPT to review your documentation
- Use github.com/kefranabg/readme-md-generator

Add awesome illustrations with Midjourney

<https://samuelberthe.substack.com/p/create-awesome-illustrations-for>



External documentation ?



samber/lo

Doc

GoDoc

Community

Changelog

💖 Sponsor



🔍 Search



K



About



Getting started



Core



Iterator



Mutable



Parallel



Glossary



Contributing



>



Getting started

Getting started

samber/lo is a Lodash-style utility library for Go 1.18+ that leverages generics to provide type-safe helper functions. The library is organized into several packages, each serving different use cases.



Install

```
go get -u github.com/samber/lo@v1
```



Core Package (lo)

The main package provides immutable utility functions for slices, maps, strings, math operations, and more. It's the core of the library with over 300+ functions.

```
import "github.com/samber/lo"
```



Install



Core Package (lo)



Iter Package (lo/it)



Mutable Package

(lo/mutable)



Parallel Package

(lo/parallel)



Key Benefits



Next Steps



Use your favorite Coding Agent.
LLM are good at generating
examples and **comments**.

Developer experience from IDE ?

```
15 /**
16  * Builder pattern implementation for creating rich error objects.
17  *
18  * The builder pattern allows for fluent, chainable error creation with contextual
19  * information. This design provides a clean API for building complex error objects
20  * while maintaining readability and flexibility.
21  *
22  * Examples:
23  *
24  * Basic error creation:
25  *   oops.Errorf("Could not fetch users: %w", err)
26  *
27  * Rich error with context:
28  *   oops.
29  *     User("steve@apple.com", "firstname", "Samuel").
30  *     Tenant("apple", "country", "us").
31  *     Errorf("403 not permitted")
32  *
33  * Error with timing and tracing:
34  *   oops.
35  *     Time(requestDate).
36  *     Duration(requestDuration).
37  *     Trace(traceID).
38  *     Errorf("Failed to execute http request")
39  *
40  * Error with custom context:
41  *   oops.
42  *     With("project_id", project.ID, "created_at", project.CreatedAt).
43  *     Errorf("Could not update settings")
44  *
45  * Thread Safety: OopsErrorBuilder instances are not thread-safe. Each builder
46  * should be used by a single goroutine. The builder methods return new instances
47  * to support chaining, which provides some isolation but doesn't guarantee
48  * thread safety for concurrent access to the same builder instance.
49  */
50
51 // OopsErrorBuilder implements the builder pattern for creating OopsError instances.
52 // It provides a fluent API for setting error attributes and creating error objects.
53 // The builder is designed to be chainable, allowing multiple method calls in sequence.
54 type OopsErrorBuilder OopsError
55
```

```
128
129 // Wrap wraps an existing error into an OopsError with the current builder's context.
130 // If the input error is nil, returns nil. Otherwise, creates a new OopsError that
131 // wraps the original error while preserving all the contextual information set
132 // in the builder.
133 //
134 // Example:
135 //
136 //   err := oops.
137 //     Code("database_error").
138 //     In("database").
139 //     Wrap(originalError)
140 func (o OopsErrorBuilder) Wrap(err error) error {
141     if err == nil {
142         return nil
143     }
144
```

```
1 package lo
2
3 // Keys creates a slice of the map keys.
4 // Play: https://go.dev/play/p/Uu11fHASqrU
5 func Keys[K comparable, V any](in ...map[K]V) []K {
6     size := 0
7     for i := range in {
8         size += len(in[i])
9     }
10    result := make([]K, 0, size)

```

```
// Shuffle returns a slice of shuffled values. Uses the Fisher-Yates shuffle.  
// Play: https://go.dev/play/p/ZTGG70UCdnp  
//  
// Deprecated: use mutable.Shuffle() instead.  
func Shuffle[T any, Slice ~[]T](collection Slice) Slice {  
    mutable.Shuffle(collection)  
    return collection  
}
```

```
to.Shuffle([]string{"foo", "bar", "", "baz"})|
```

Index your documentation

DeepWiki

Index your code with  Devin

 Share



Which repo would you like to understand?

 Search for repositories (or paste a link)

+
Add repo

microsoft / vscode
Visual Studio Code

☆ 170.1k

antiwork / gumroad

☆ 5.2k



Context7

Personal  [Dashboard](#)

[MCP Server](#)

+ Add

Add Libraries

Select and add your documentation source



GitHub



GitLab



LLMs.txt



Website

☐ Private Repo

GitHub URL

Use public/open-source GitHub repos with URLs like: <https://github.com/username/repo>

Submit

Last indexed: 14 October 2025 (8fd70e)

Overview

Getting Started

Core Library

Observable and Observer Patterns

Subjects

Core Operators

Testing Observables

Core Testing Framework

Testify Plugin

Plugin Ecosystem

Data Transformation Plugins

Template Plugin

Strconv Plugin

Sort Plugin

Regexp Plugin

Iter Plugin

I/O and Communication Plugins

WebSocket Client Plugin

Overview

> Relevant source files

This document provides an architectural overview of the **ro** reactive programming library for Go. It covers the repository structure, core abstractions, plugin ecosystem, testing infrastructure, and development workflows. For hands-on tutorials and basic usage, see [Getting Started](#). For in-depth technical details of the core library, see [Core Library](#).

Purpose and Scope

ro is a reactive programming library for Go that implements the Observable pattern with a comprehensive plugin ecosystem. This overview explains:

- The repository's monorepo structure and Go workspace organization
- Core reactive primitives (Observable, Observer, Subject)
- The 23+ plugin modules that extend the core library
- Testing frameworks for validating observable behavior
- CI/CD pipelines and release automation
- Enterprise Edition features

For specific plugin documentation, see [Plugin Ecosystem](#). For testing observables, see [Testing Observables](#). For contributing and development workflows, see [CI/CD and Development](#).

Repository Structure

Refresh this wiki



Enter email to refresh

On this page

Overview

Purpose and Scope

Repository Structure

Workspace Organization

File Structure

Core Library Architecture

Core Reactive Primitives

Plugin Ecosystem

Plugin Categories and Module Paths

Plugin Version Requirements

Testing Infrastructure

Testing Framework Components

Testing Workflow

CI/CD and Development Workflow

GitHub Actions Workflows

Test Matrix Coverage



SEO is dead, long live GEO

Markdown everywhere

Short sentences

FAQ

Mermaid diagrams

llms.txt

...

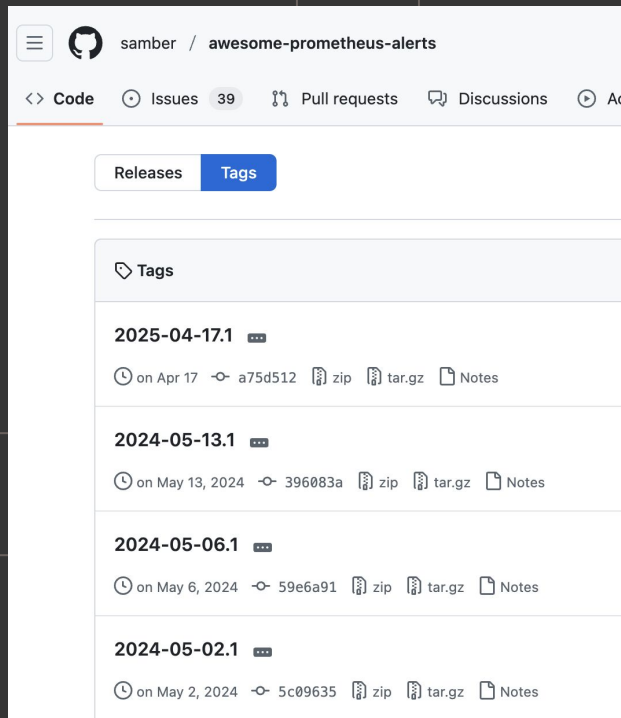
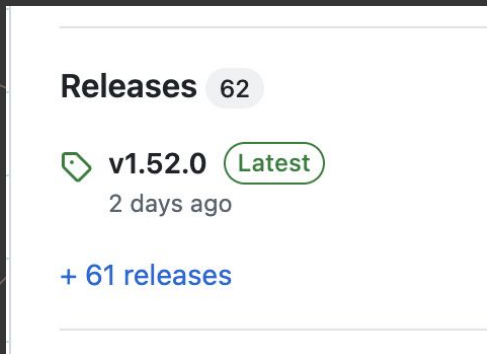


Keep backward compatibility and
long-term support (LTS)

Cross-platform compatibility

Releases

- Stick to SEMVER strongly
- Avoid breaking changes
- Releases often
- Generate releases in a Github Action



Write changelogs

v1.52.0

Latest

Compare



github-actions released this yesterday · 1 commit to master since this release · v1.52.0 · a25c42d

Thrilled to announce the latest release of `samber/lo` !

This version brings major improvements:

- A new `it/` sub-package for everything related to the `iter` package added in Go 1.23. Big up to [@NathanBaulch](#) for its help on this
- A new documentation -> lo.samber.dev
- Lots of micro fixes and performance improvements
- Better code quality with many more linters

🌟 A good introduction to streams in Go !



What's Changed

- fix(chunk): Copy chunk in a new slice by [@samber](#) in [#648](#)
- fix(zipx): fix the max length calculation by [@samber](#) in [#649](#)
- fix(retry): fix potential race condition in debounce callback by [@samber](#) in [#650](#)
- feature: add Mode function with tests and documentation by [@ghosx](#) in [#644](#)
- add FilterKeys for map by [@rasha108bik](#) in [#631](#)
- feat: adding FilterValues and fix FilterKeys signature by [@samber](#) in [#661](#)

Write changelogs

[Releases](#) / Edit v1.52.0

Edit v1.52.0

 Tag: v1.52.0 ▼

✓ Existing tag

Release title

v1.52.0

Release notes

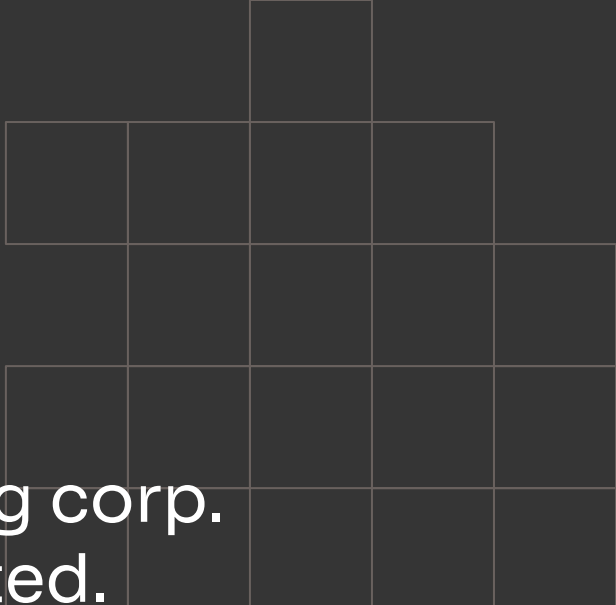
Select a previous tag to create generated release notes

 Previous tag: v1.51.0 ▼

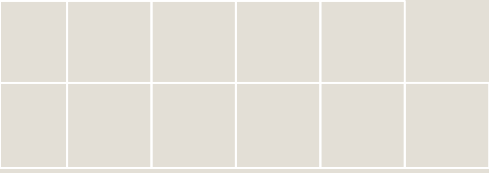
Generate release notes

Write

Preview

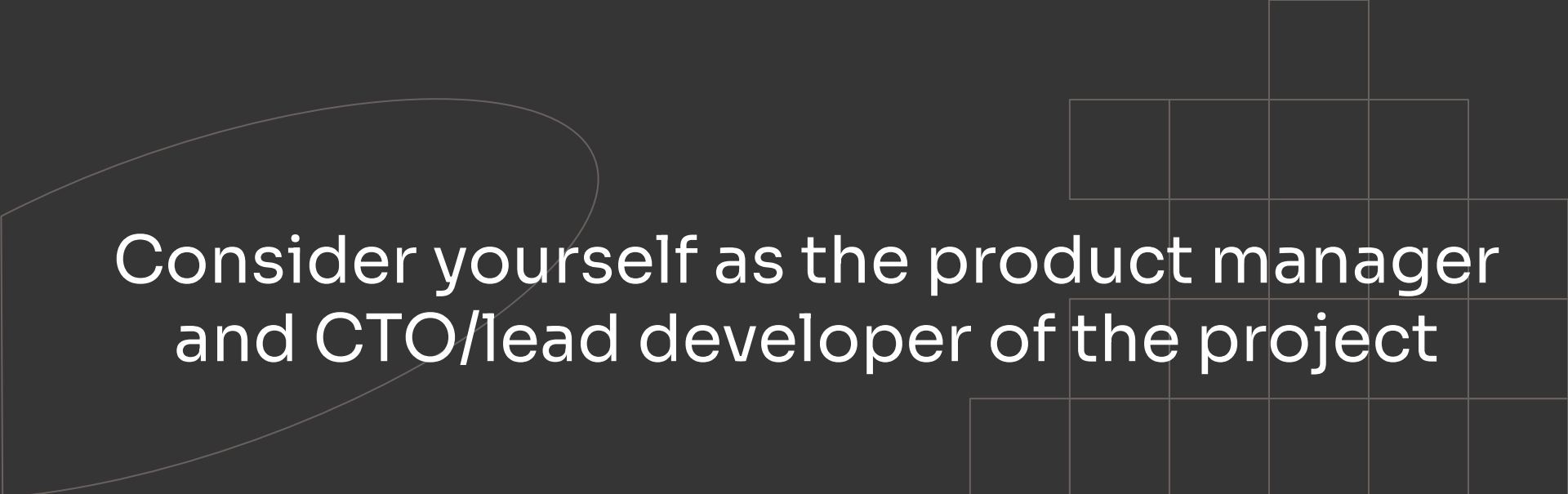


Releases are reviewed by big corp.
Missing commits are detected.



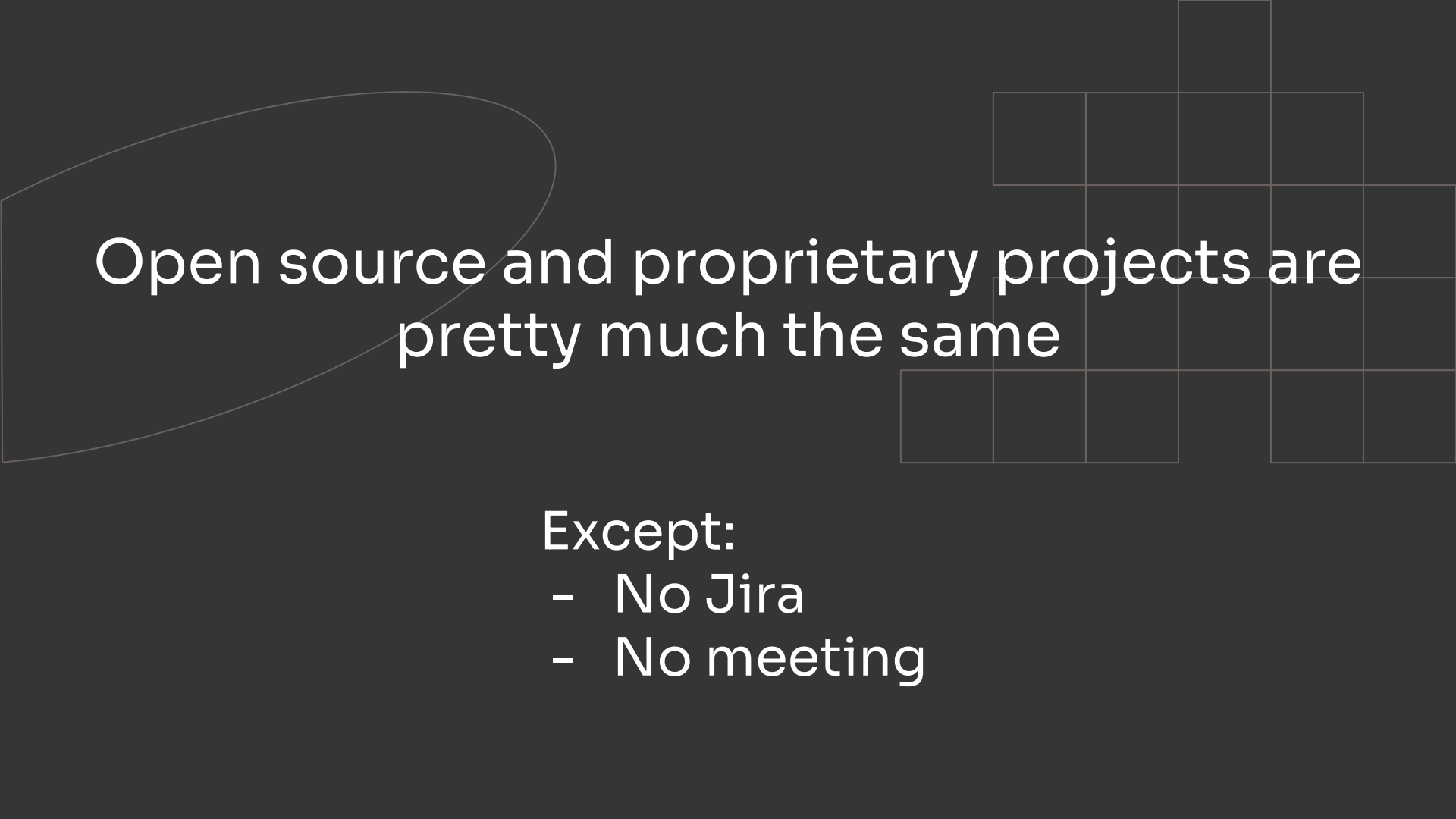
3- Take care of contributor experience





Consider yourself as the product manager
and CTO/lead developer of the project

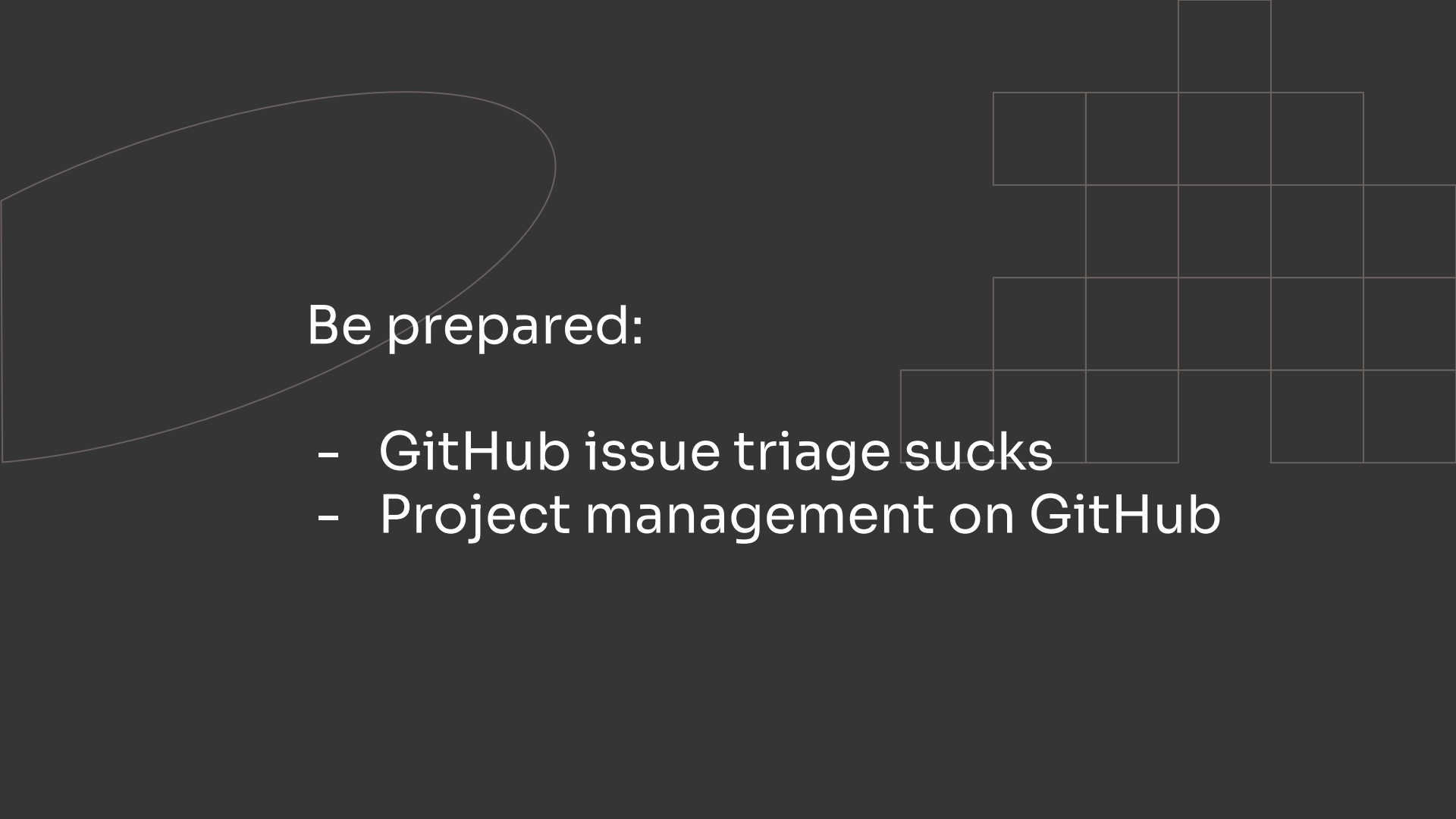
Organize the community
and the discussions



Open source and proprietary projects are
pretty much the same

Except:

- No Jira
- No meeting

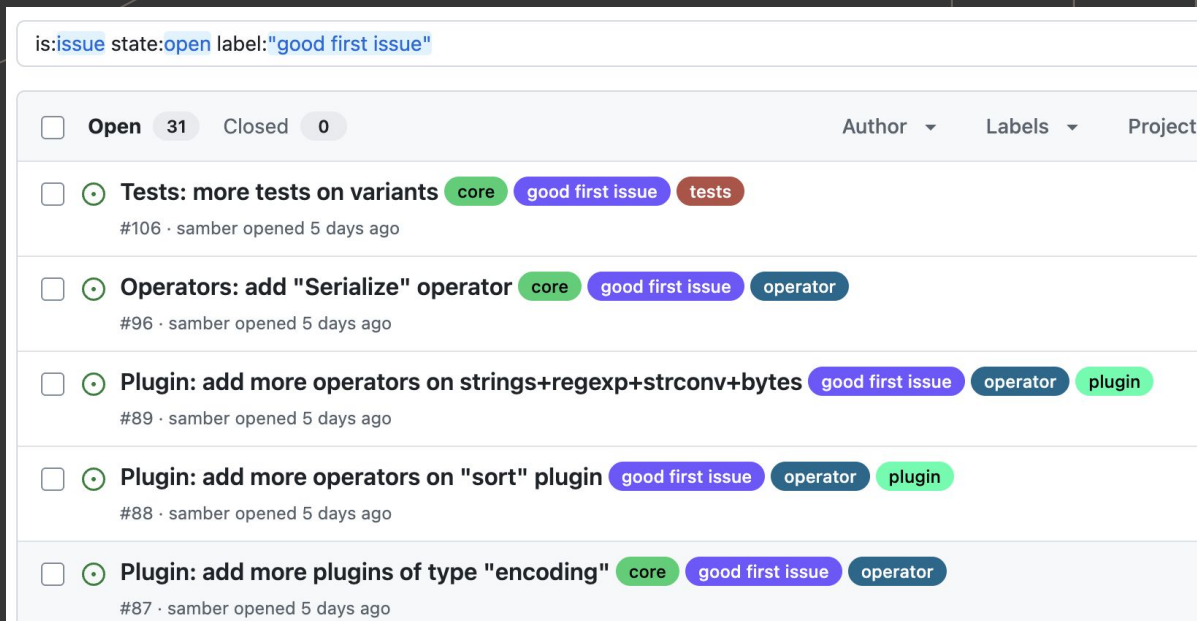


Be prepared:

- GitHub issue triage sucks
- Project management on GitHub

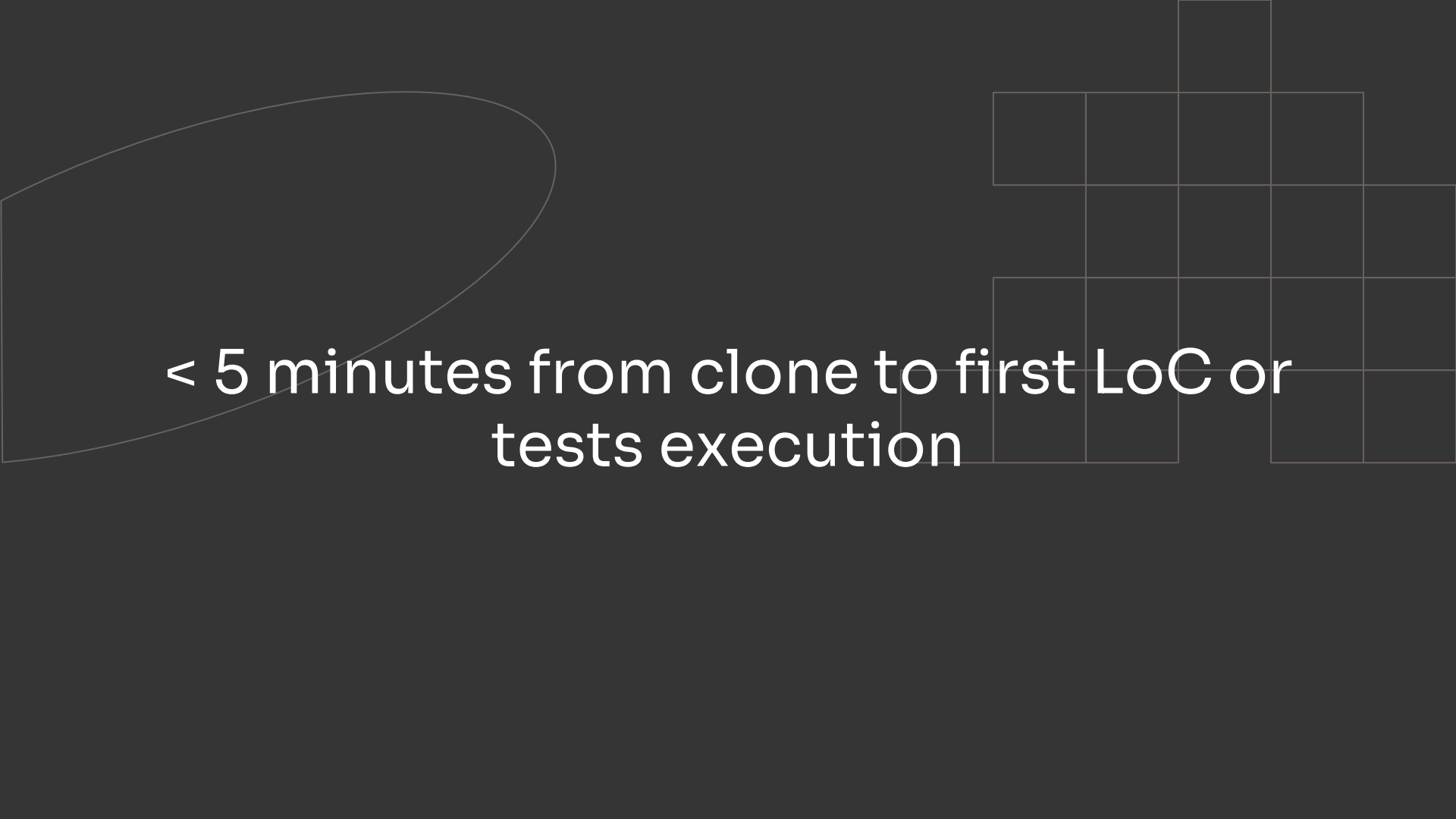
Focus yourself on hard tasks and reviews

...and redirect new contributors to “*good first issues*”



The screenshot shows a GitHub interface with a search bar at the top containing the text "is:issue state:open label:\"good first issue\"". Below the search bar, there are filters for "Open" (31) and "Closed" (0). The main content area displays a list of five issues, each with a checkbox, a green circle icon, a title, and several labels. The issues are:

- ☐ **Tests: more tests on variants** (core, good first issue, tests) #106 · samber opened 5 days ago
- ☐ **Operators: add "Serialize" operator** (core, good first issue, operator) #96 · samber opened 5 days ago
- ☐ **Plugin: add more operators on strings+regexp+strconv+bytes** (good first issue, operator, plugin) #89 · samber opened 5 days ago
- ☐ **Plugin: add more operators on "sort" plugin** (good first issue, operator, plugin) #88 · samber opened 5 days ago
- ☐ **Plugin: add more plugins of type "encoding"** (core, good first issue, operator) #87 · samber opened 5 days ago



< 5 minutes from clone to first LoC or
tests execution

Write a contributing guide (CONTRIBUTING.md)



Contributing

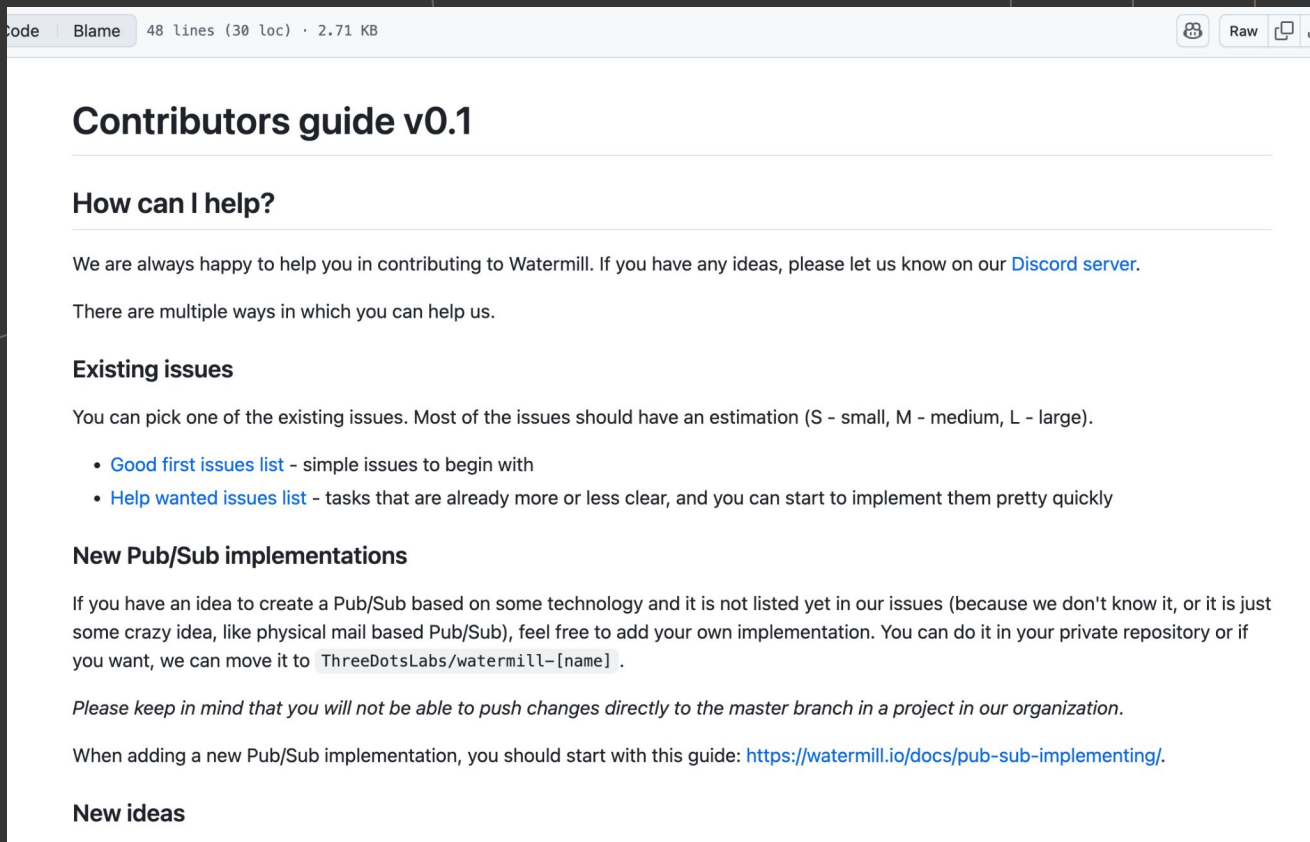
- Ping me on Twitter [@samuelberthe](#) (DMs, mentions, whatever :))
- Fork the [project](#)
- Fix [open issues](#) or request new features

Don't hesitate ;)

```
# Install some dev dependencies  
make tools
```

```
# Run tests  
make test  
# or  
make watch-test
```

Write a contributing guide (CONTRIBUTING.md)



[👋 About](#)[🚀 Getting started](#)[🔑 Core](#)[🔄 Iterator](#)[🔗 Mutable](#)[🏗️ Parallel](#)[📖 Glossary](#)[👉 Contributing](#)[👉 Contributing](#)

Contributing

Hey! We are happy to have you as a new contributor. 🙌

For your first contribution please follow some guidelines:

Function Naming

Function naming: helpers must be self-explanatory and respect standards (other languages, libraries...). Feel free to suggest many names in your contributions.

Variadic functions

Many functions accept variadic parameters (like `Lo.Keys(...map [K]V)` accepting multiple maps), providing flexibility while maintaining type safety.

Slice type Parameters

Functions use `~[T]` constraints to accept any slice type, including named slice types, not just `[T]`. This design choice makes the library more flexible in real-world usage.

Variants

When applicable, some functions can be added to sub-package as well: `mutable`, `it` and `parallel`. Add a documentation for each helper.

[Function Naming](#)[Variadic functions](#)[Slice type Parameters](#)[Variants](#)[Testing](#)[Benchmark and performance](#)[Documentation](#)[Examples](#)

Write a contributing guide



👁️ **claude** bot reviewed 5 days ago

operator_utility.go

```
634 + // Serialize ensures thread-safe message passing by wrapping any observab
635 + func Serialize[T any]() func(Observable[T]) Observable[T] {
636 +     return func(source Observable[T]) Observable[T] {
637 +         return NewSafeObservableWithContext(func(subscriberCtx context.Cor
```



claude bot 5 days ago

The current implementation doesn't properly handle context cancellation. You should pass `source.SubscribeWithContext()` call, not the context from the wrapper function. The current might break context propagation.



Reply...

Unresolve conversation

samber marked this conversation as resolved.

```
- uses: anthropics/claude-code-action@v1
env:
  ANTHROPIC_BASE_URL: ${ secrets.ANTHROPIC_BASE_URL }
  ANTHROPIC_MODEL: ${ secrets.ANTHROPIC_MODEL }
  ANTHROPIC_AUTH_TOKEN: ${ secrets.ANTHROPIC_AUTH_TOKEN }
with:
  anthropic_api_key: ${ secrets.ANTHROPIC_AUTH_TOKEN }
prompt: |
  REPO: ${ github.repository }
  PR NUMBER: ${ github.event.pull_request.number }
  AUTHOR: ${ github.event.pull_request.user.login }
```

Please review this pull request with a focus on:

- Code quality and best practices
- Potential bugs or issues
- Security implications
- Performance considerations
- Potential breaking changes
- Developer experience and documentation

Check contribution guidelines: `./docs/docs/contributing.md`

Check extending ro guidelines: `./docs/docs/hacking.md`

Check examples/documentation guidelines: `./docs/CLAUDE.md`

Check troubleshooting guidelines: `./docs/docs/troubleshooting/`

If you need more context on the project, read the LLMs documentation: `./docs/static/llms.txt`

Issue/PR templates

Create new issue 

 **Bug Report** 
Create a report to help us improve.

 **Feature request** 
Suggest an idea for this project.

Report a security vulnerability 
Please review our security policy for more details

 **Documentation** 
Please take a look to our documentation.

 **Questions** 
If you have a question, or are looking for advice, please post on our Discussions forum!

 **Chat on Slack** 
Maybe chatting with the community can help

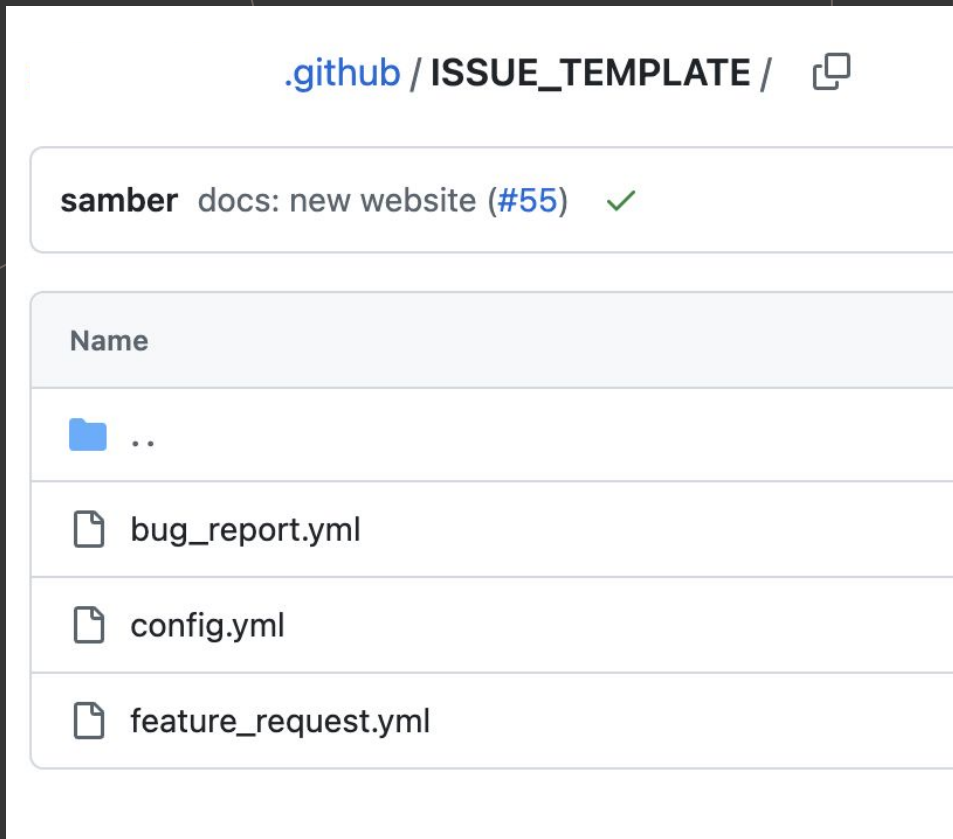
Issue/PR templates

Ask for reproducible demo of the bugs.

Write checklist for new features

...or you're going to waste massive amount of time
with junior developers.

Issue/PR templates



Labels

Search all labels

12 labels

bug	Something isn't working
dependencies	Pull requests that update a dependency file
discussion needed	Further information is requested
enhancement	New feature or request
go	Pull requests that update go code
good first issue	Good for newcomers
help wanted	Extra attention is needed
javascript	Pull requests that update javascript code
L	large issue
M	medium issue
New Pub/Sub	

Labels

Milestone

  **samber** / **do**

Type  to search

  |     

<> Code • Issues 27 • Pull requests 5 • Discussions • Actions • Security 25 • Insights • Settings

Milestones

New milestone

Open 3 Closed 0

Sort

v2.2
No due date • 0/2 issues closed

0% complete 2 open 0 closed

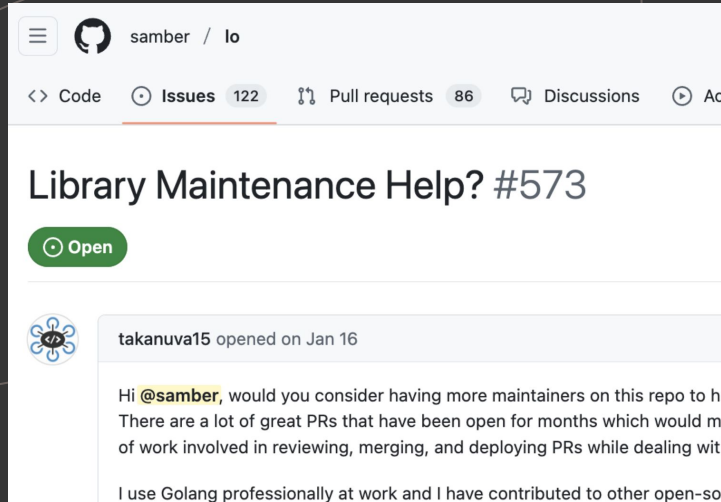
v2.1
No due date • 2/21 issues closed

9% complete 19 open 2 closed

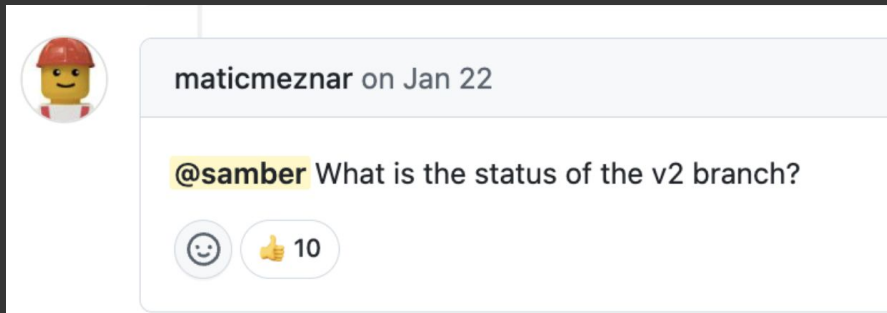
v2
No due date • 9/12 issues closed

75% complete 3 open 9 closed

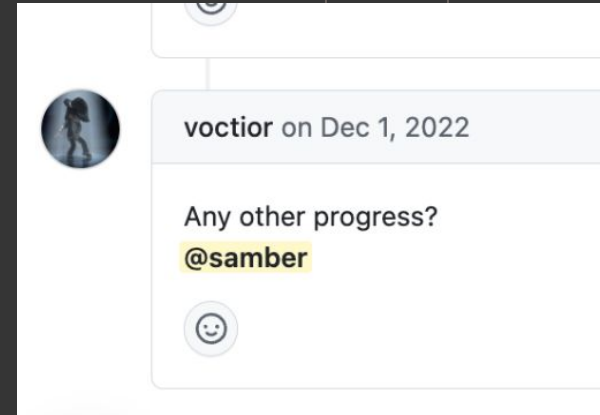
Be diligent



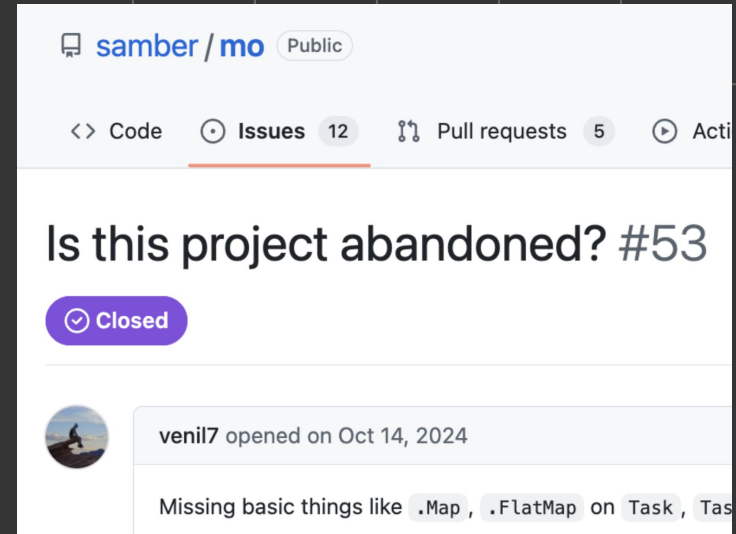
GitHub interface for the repository `samber / io`. The navigation bar shows `<> Code`, `Issues 122` (highlighted), `Pull requests 86`, and `Discussions`. The issue title is **Library Maintenance Help? #573**, with a green `Open` button. The issue was opened by `takanuva15` on Jan 16. The content shows a message to `@samber` asking about adding more maintainers to the repository.



GitHub comment by `maticmeznar` on Jan 22. The comment asks `@samber` about the status of the v2 branch. It has 10 thumbs up and a smiley face reaction.



GitHub comment by `vooctior` on Dec 1, 2022. The comment asks "Any other progress?" and mentions `@samber`. It has a smiley face reaction.



GitHub interface for the repository `samber / mo`, marked as `Public`. The navigation bar shows `<> Code`, `Issues 12` (highlighted), `Pull requests 5`, and `Actions`. The issue title is **Is this project abandoned? #53**, with a purple `Closed` button. The issue was opened by `venil7` on Oct 14, 2024. The content shows a message about missing basic things like `.Map`, `.FlatMap` on `Task`, `Tas`.

Project management is
painful on GitHub

 Neat

To do



181

Mentioned in lo

4d

Allow Union/Intersect to take many lists



Subscribed by willie68  in do

6d

Needed InvokeNamedAs, MustInvokeNamed...



Comment by edocevol  in do

24d

Proposal (v2): Injector should implement co...



Subscribed by Vilsol  in do

24d

feat: add InvokeAsAll for bulk service invoca...



Subscribed by james-at-upmind  in do

25d

Access to context.Context in Provider & 'mu...



Subscribed by ohroy  in do

25d

feat: add Delete method to remove scopes



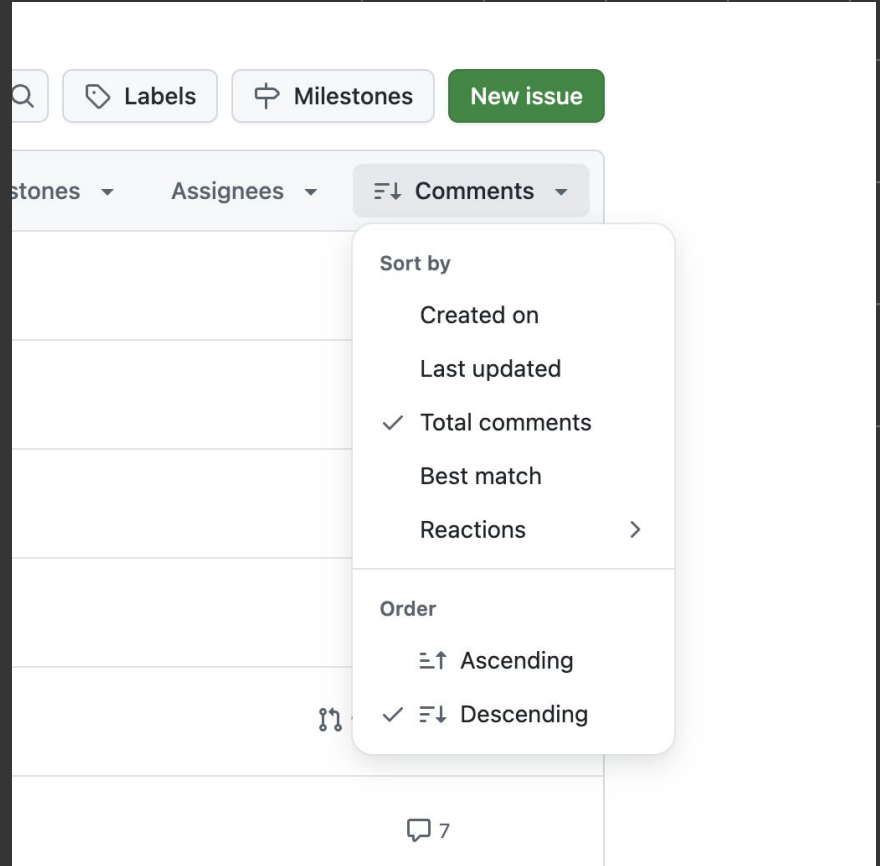
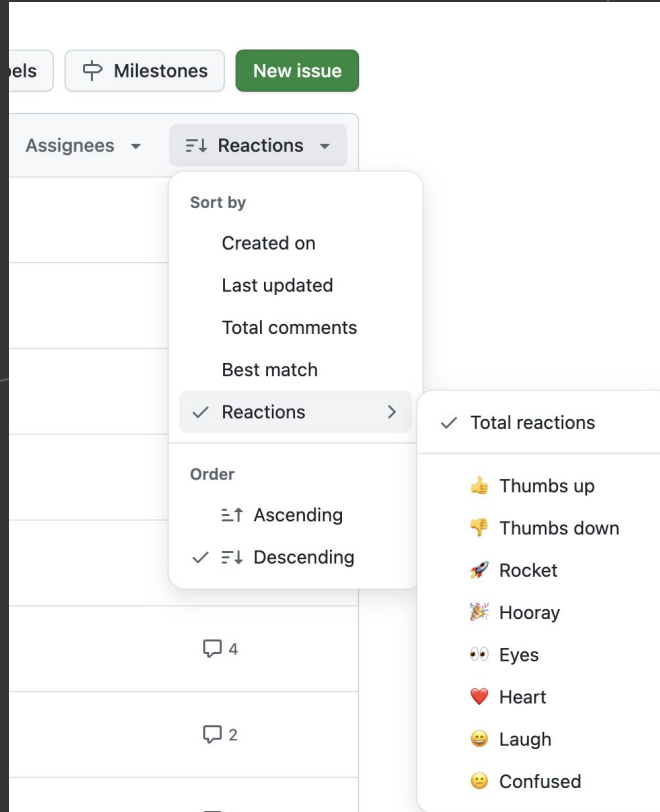
Subscribed by ohrov  in do

25d

Neat



Find hot issues/PR



Issue triage: deduplication, labeling, reformulate...

Multiple registration and bulk invocation #114

Open



samber opened on Aug 18 · edited by samber

Currently, the framework authorizes only a single registration per type. Calling twice `do.Pro` trigger an error. In that situation, the invocation is much simpler because we can find at most

However, starting with v2, we will support service aliasing, either via explicit declaration (`do` implicit invocation (`do.InvokeAs [Alias] (...)`), which loads the first service implementing t

Note: currently, a developer can override a service using `do.OverrideXXX(...)`. It has been replacement in unit tests, and should be combined with `.Clone()` method of `RootScope`. If service invocation, the service is replaced, but not unloaded/shutdowned.

Multiple contributors asked for an improvement: [#33](#) [#75](#) [#45](#) [#52](#) [#117](#)

Poke [@matdurand](#) [@svenwltr](#) [@danilobuerger](#) [@d-enk](#) [@mbark](#) [@GregPja](#) [@Jictyvoo](#)

Don't be an asshole with contributors...

... but some contributors will be an asshole with you



Emily Kager @EmilyKager · Aug 19, 2021

working in open source is the working retail of software



Wardell Bagby

@wardellbagby



- entitled people screaming at you to do stuff
- being paid extremely little
- large companies taking advantage of you.

You hit the damn nail RIGHT ON THE HEAD.

Say “no”



Solomon Hykes 

@solomonstre

Rule #1 of open-source: no is temporary, yes is forever.

10:38 PM · Mar 30, 2016

Closed issue -> no problem 🤔



github-actions bot commented now



This issue has been automatically locked since there has not been any recent activity after it was closed.
Please open a new issue for related bugs.



github-actions bot locked as **resolved** and limited conversation to collaborators now

Hard to collect feedback as a maintainer

Spec: about index argument in iteratee #579

[Edit](#)[New issue](#)[Open](#)

samber opened on Jan 25

Owner ...

Go does not support optional arguments. We have to duplicate helpers if we want to make the `index int` argument optional.

I'm opening this issue to discuss a new standard for naming helpers.

- `lo.Map(list, func(item T) U -> no index`
- `lo.MapI(list, func(item T, index int) U -> index`

The `I` suffix might not be comprehensible. I would suggest `WithIndex` as second choice.

It might introduce a large breaking change because `Map` and `Filter` are concerned. (see discussion about v2: [#578](#))

Also, I wonder what would be the best type for this index:

- `int` is simple
- `int64` for larger collections (useless 99.9% of time)
- `uint64` the most accurate type

Don't forget that passing a function to a helpers is a very developer-friendly API: `lo.Map(collection, aFunc)`. Since `int` is the most used type it seems valuable to avoid any exotic typing and cast.

Linking issues & PR: [#565](#)

Create sub-issue



7



6

Assignees



No one - [Assign yourself](#)

Assign to Copilot

Labels



breaking change **spec**

Projects



No projects

Milestone



☐ **v2**

No due date

Relationships



None yet

Development

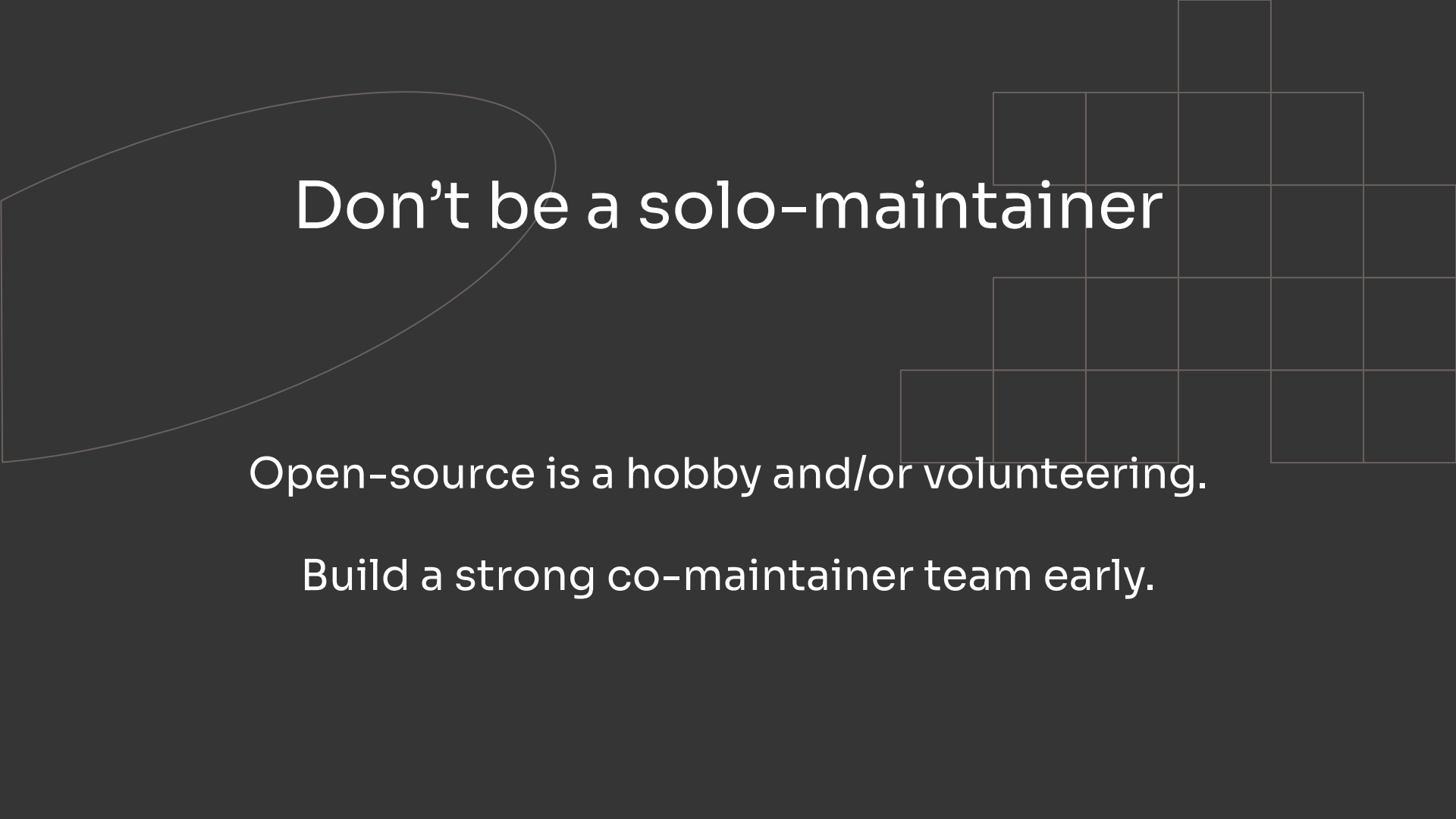


Code with agent mode

[Create a branch](#) for this issue or link a pull request.

Notifications

Customize



Don't be a solo-maintainer

Open-source is a hobby and/or volunteering.

Build a strong co-maintainer team early.

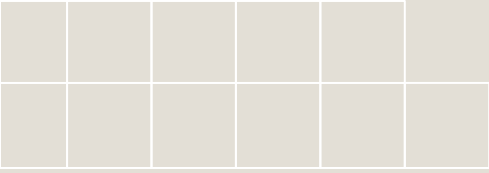
Don't be a solo-maintainer

Find key contributors on the front page of your project:

Contributors 147



+ 133 contributors



4- Spread the word*

and get adoption

*Feel free to yell at me during the next 5min

Snowball effect

more users



more contributors



better project



repeat

Imagine you need a **React date picker**.

You go on **GitHub** and find **4** different open-source projects that could fit.

Turn to the persons next to you, and tell them your criteria (1 min)

blabla / datepicker

Q Type / to search

datepicker Public

Sponsor Watch 4 Fork 22 Star 335

main Go to file Code

Elon chore: bump react to the v19 (#75) a23eebc · 2 years ago 35 Commits

.github fix: support node 16 for legacy rea... 2 years ago

.husky infra: move to pnpm workspace (#... 4 years ago

About
Date picker

 Readme

wojtekmaaj / react-date-picker

Q Type / to search

10 14 Discussions Actions Wiki Security Insights

react-date-picker Public

Sponsor Watch 11 Fork 202 Star 1.3k

main Go to file Code

wojtekmaj Bump typescript from 5.5.2 to ... d92806f · 2 weeks ago 1,159 Commits

.github Remove unnecessary provenance f... 2 weeks ago

.husky Replace ESLint and Prettier with Bi... last year

About
A date picker for your React app.
 projects.wojtekmaj.pl/react-date-picker

Used by 18.9k



Contributors 147



+ 133 contributors

Unwatch

78



Fork

900



Star

20.3k

<> Code

About



🔥 A Lodash-style Go library based on Go 1.18+ Generics (map, filter, contains, find...)

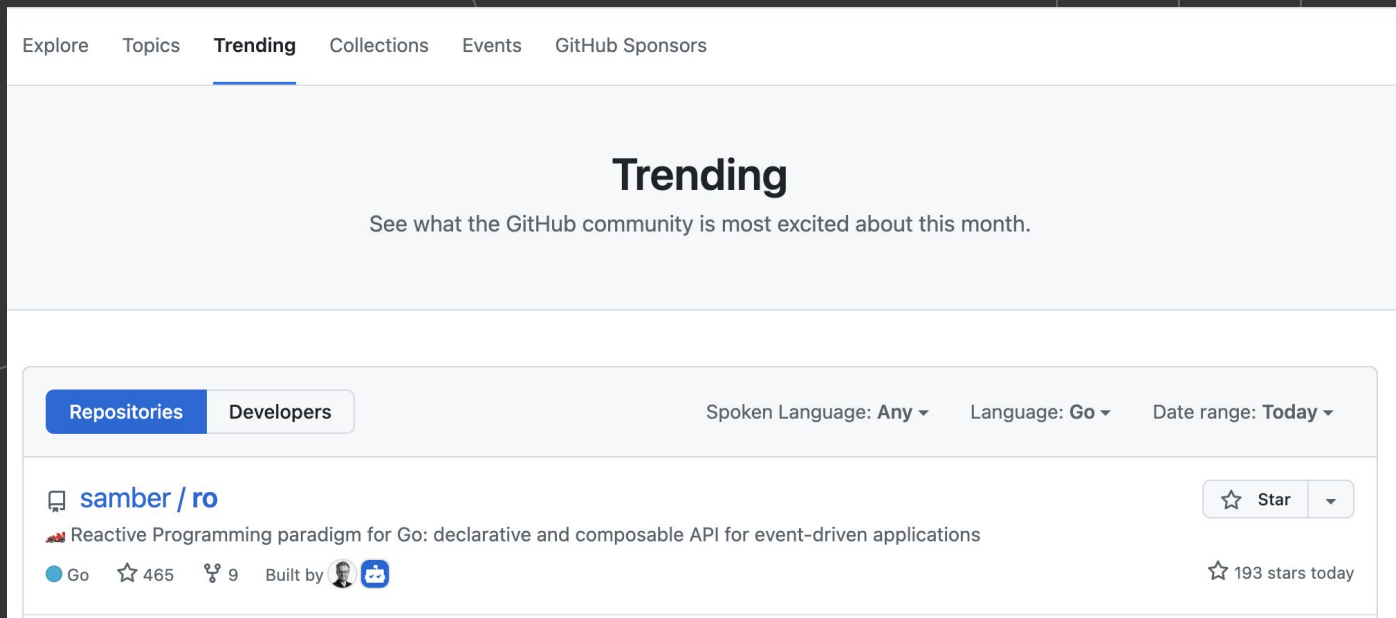


682 Commits

3 days ago

Social proof is powerful

Showcase your project



Organize an artificial launch with friends and family
and reach 100 stars, 1k+ views and 10+ forks ASAP,
then go organic.

Snowball (GitHub home page)

Home

Ask Copilot



Create a profile README for me

<> Make a Pong game

<> Get code feedback

Feed

Filter 3



github/spec-kit released

1 hour ago



Spec Kit Templates - 0.0.73

This is the latest set of releases that you can use with your agent of choice. We recommend using the Specify CLI to scaffold your projects, however you can download these independently and manage ...

Read more



1

Snowball (GitHub home page)

See also:

- [samber/lo](#): A Lodash-style Go library based on Go 1.18+ Generics
- [samber/do](#): A dependency injection toolkit based on Go 1.18+ Generics
- [samber/mo](#): Monads based on Go 1.18+ Generics (Option, Result, Either...)



tbourelly starred 2 of your repositories

yesterday



samber/mo



Star



Monads and popular FP abstractions, powered by Go 1.18+ Generics (Option, Result, Either...)



Go



3.2k



samber/ro



Star



Reactive Programming paradigm for Go: declarative and composable API for event-driven applications



Go



305

Showcase your project

- Reddit (in multiple sub-reddits) / V2EX
- HackerNews
- Social networks: X, Bsky, LinkedIn
- Slack/Discord communities
- daily.dev
- Get featured in newsletter
- Awesome lists
- Post comments on blog posts (medium, substack, etc.)
- Make a list of promoters (friends, contributors, etc...)

Follow the platform's rules in order to reach lots of people

Talk to meetups, podcast and conferences

- Visibility on social media
- Youtube video
- Costly

<https://developers.events>



Content marketing

Write and publish technical posts

~~Medium~~, Substack, Hackernoon or dev.to

Use ChatGPT+Midjourney

Add a call-to-action

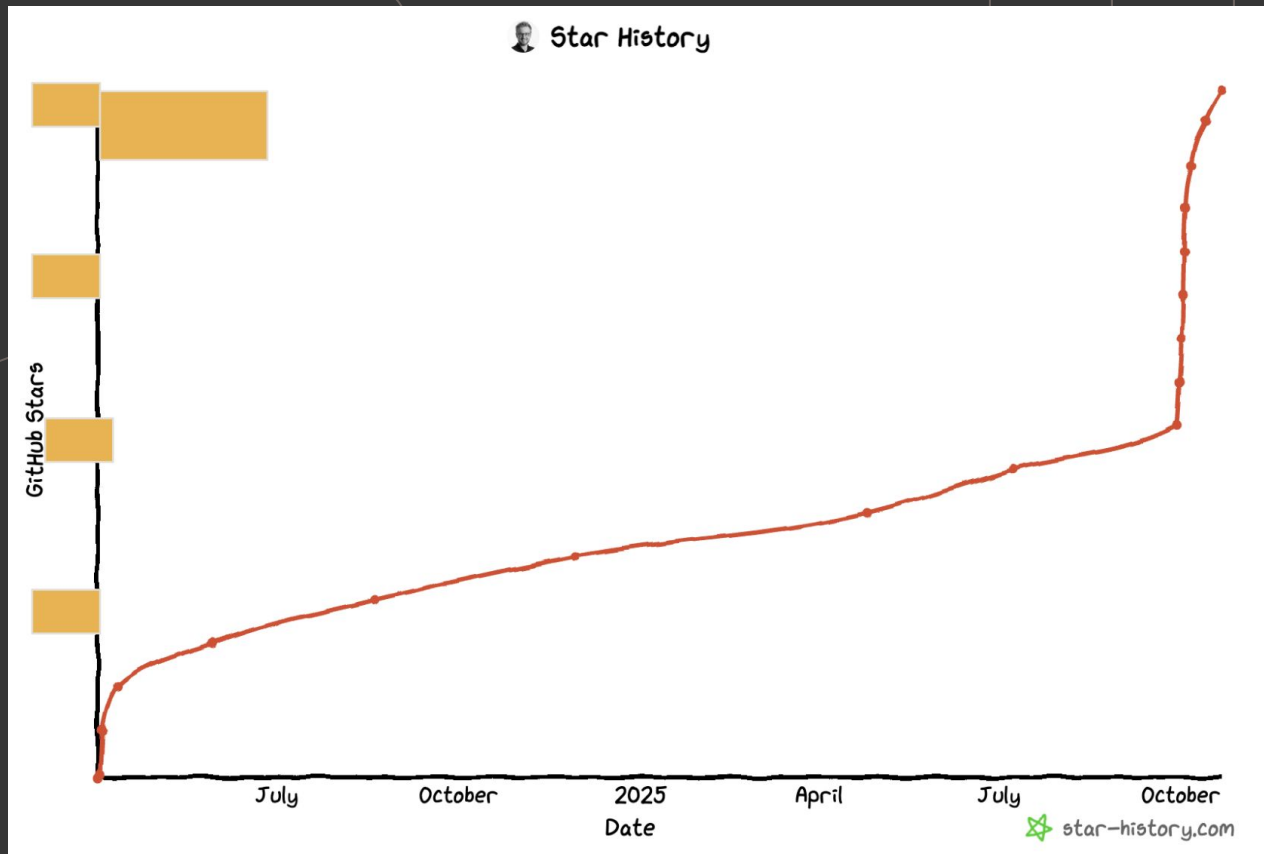
Want to try? Start here: github.com/samber/ro.

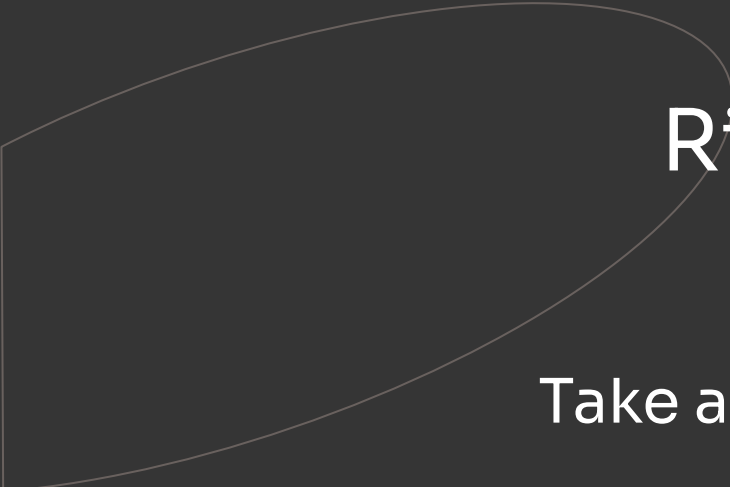
Thanks!

If you enjoy my work, consider sponsoring me on GitHub. Your support helps me keep blogging and coding open-source projects 🙌 github.com/sponsors/samber

<https://samuelberthe.substack.com/p/go-beyond-goroutines-introducing>

I wrote 2 posts on Substack (end of september 2025)





Ride the wave

Take advantage of the hype
(currently: AI)

Code examples, blog posts...

Outbound marketing

```
> git clone <similar-projects>  
> git log --format='%aN,%aE' | sort -u > contributors.csv
```

```
github.com/samber/lo master x ▶ git log --format='%aN <%aE>' | sort -u | grep -v users.noreply.github.com
```

```
Adam Szaraniec <adam.szaraniec@wahed.com>  
Adam Szaraniec <adam@wahedinvest.com>  
Adrian <colid.bondzio@gmail.com>  
Affan Shahid <affan.shahid.94@gmail.com>  
Albert Salim <albertlimca@gmail.com>  
Alex Efros <powerman@powerman.name>  
Alex Klepov <alexezm@yandex.ru>  
Alexandre Lamarre <alexandre.lamarre@suse.com>  
Ali Sunjaya <ali.sunjaya@fazzfinancial.com>  
Ali Sunjaya <ali.sunjayaa@gmail.com>  
Anatoly Ibragimov <metalex100@gmail.com>  
Anatoly Rugalev <anatoly.rugalev@gmail.com>  
Andre Soares <andregsilv@gmail.com>  
apriil15 <nlwkobe30@gmail.com>  
Aria <aria3ppp@gmail.com>  
Artem Mikheev <artemmikheev1311@gmail.com>
```



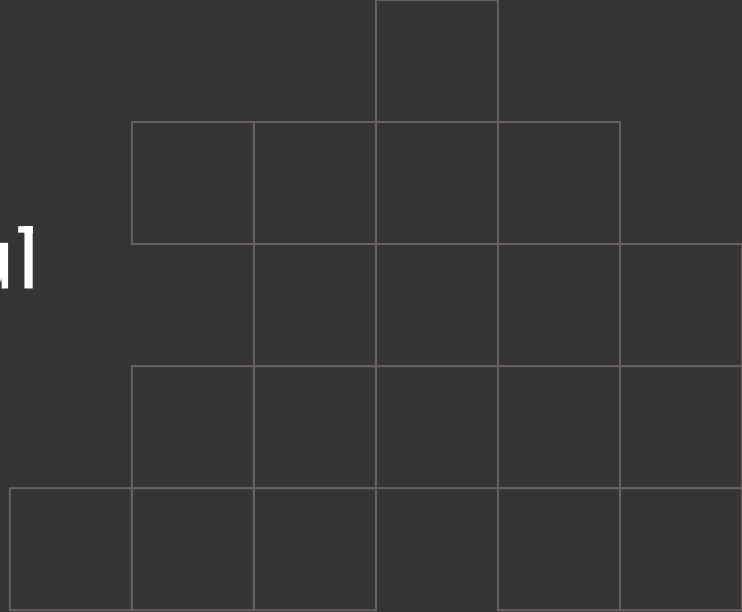
Be useful



Don't spam



Share something cool

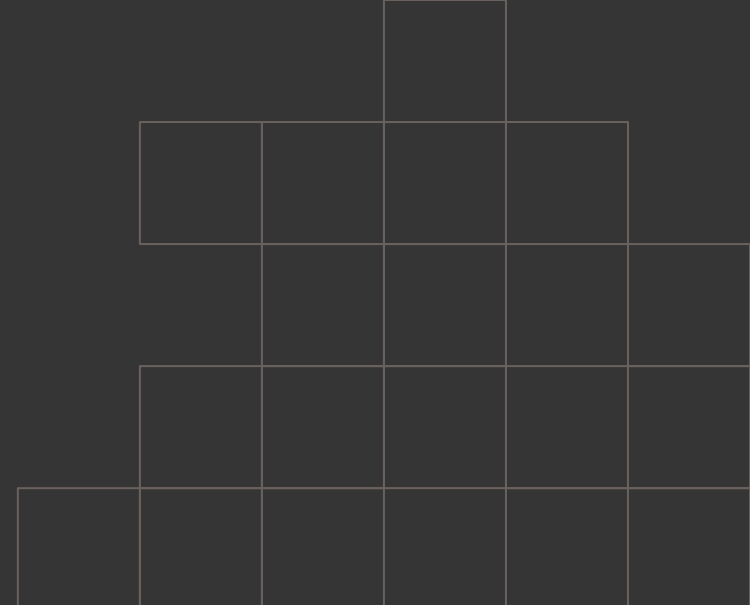




Listen:

- Reddit
- HackerNews
- (Lobster?)
- Google Alert
- X, Bsky, LinkedIn

...and enter the conversation



alerthn.com



[Home](#) [FAQ](#) [Dashboard](#) [Account](#) [About](#) [Logout](#)

F5Bot is a free service that emails you when your selected keywords are mentioned on [Reddit](#), [Hacker News](#), or [Lobsters](#). Use it to monitor your brand, your projects, or just topics that you're interested in.



Alert HN

My Filters

+ New Filter

Settings

Logout

Live email alerts for Hacker News

Free email notifications for comments and posts matching custom queries.

Get notified in time to participate in the discussion!

Sign up

Log in

f5bot.com

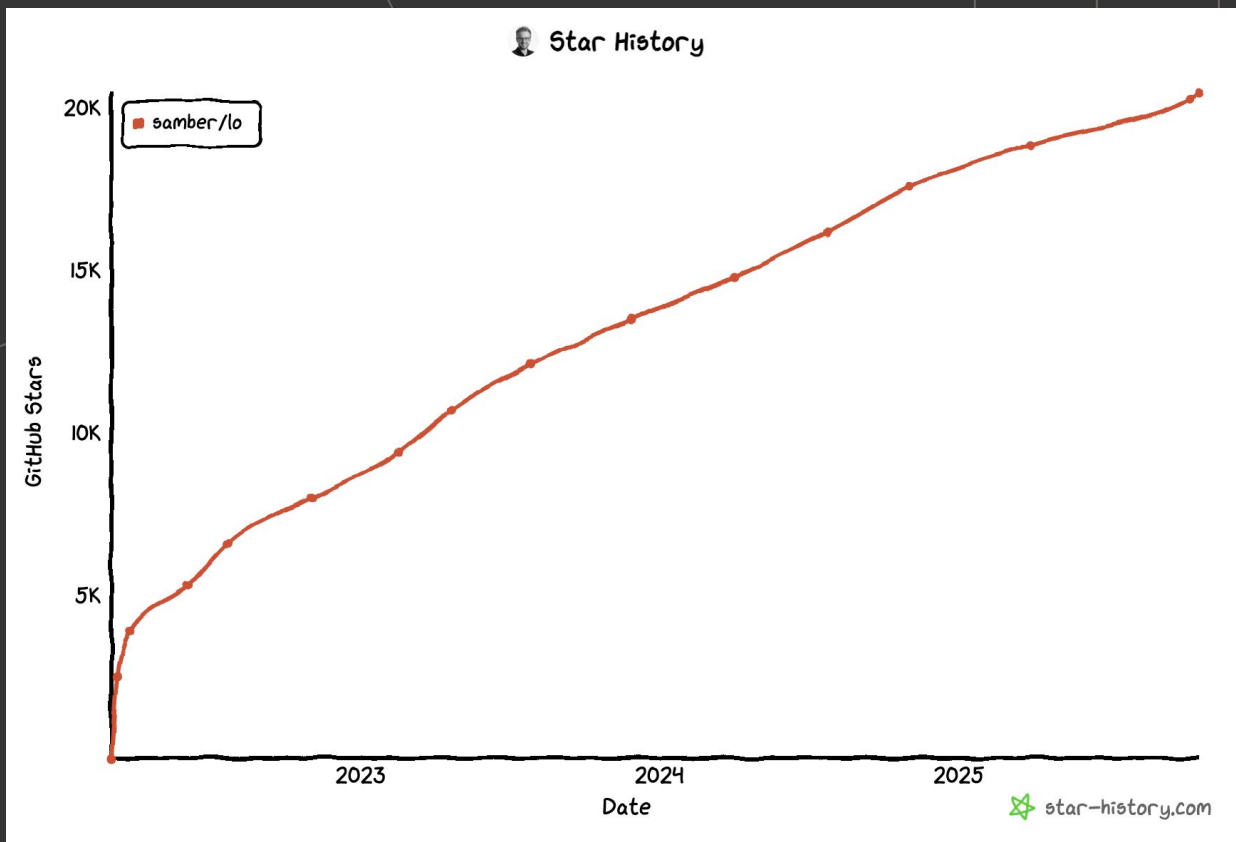
Stars can teach you your target audience

Companies, country,
job title...

Stargazers

All 310 You know 3

 burahimu Bedrock Streaming Follow	 ivanopcode @wildberries-tech Follow	 niel-biswas Poland Follow
 chaporgin @neondatabase Follow	 jferri @Agentero Follow	 Myzel394 127.0.0.1 Follow
 hslatman @smallstep Follow	 avary Istanbul Follow	 owais @splunk Follow
 grinish21 Joined on Oct 12, 2015 Follow	 Batnaidan Chimege Follow	 ikarius Portail RSE - DORA / GIP de l'Inclusion Follow
 kalifs MeisterTechnology Follow	 SaiNageswarS Microsoft Follow	 broem University Follow
 markusressel Adesso Mobile Solutions Follow	 tbourely Lyon, FR Follow	 oanogin Ottawa Follow



Git clones



Visitors



github.com/samber/*****/graphs/traffic

Image Window

tes

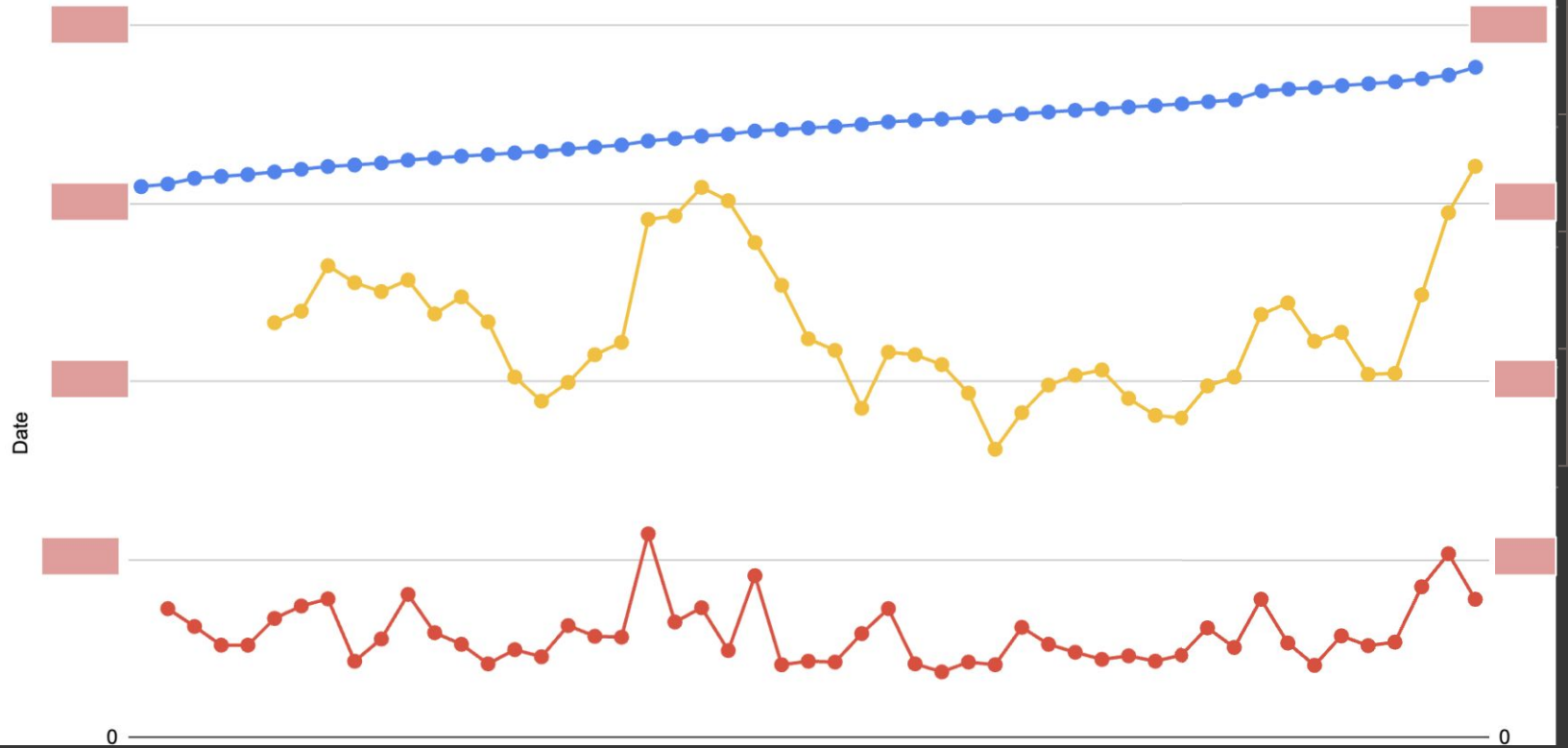
Site	Views	Unique visitors
 Google	2,926	1,885
 github.com	764	431
 stackoverflow.com	250	200
 hub.github.com	204	100
 t.co	156	92
 qiita.com	106	76
 dev.classmethod.jp	36	28
 DuckDuckGo	30	21
 Bing	24	23
 news.ycombinator.com	21	19

Popular content

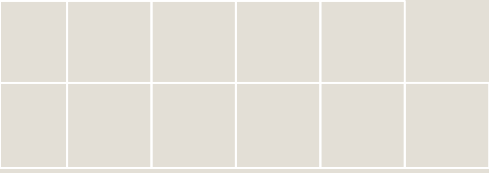
Content	Views	Unique visitors
 github/hub	3,635	2,497
 Issues	149	106
 Error creating pull request: Unproce...	101	74
 Evaluate embedded environment v...	98	72
 Pull Requests	79	32
 cloning private repositories · Issue ...	73	65
 hub/etc/hub.zsh_completion at master	71	51
 hub/features/zsh_completion.featur...	69	60
 hub/lib/hub at master	68	43
 hub/lib at master	67	45

github.com/samber/****/graphs/traffic

Total Stars + Stars / week + Stars / month



http://docs.google.com/spreadsheets/d/1lCY63TGxvyRfI_A_QdI_we5YA6VuZUcB6yuFYCYHMXw/edit

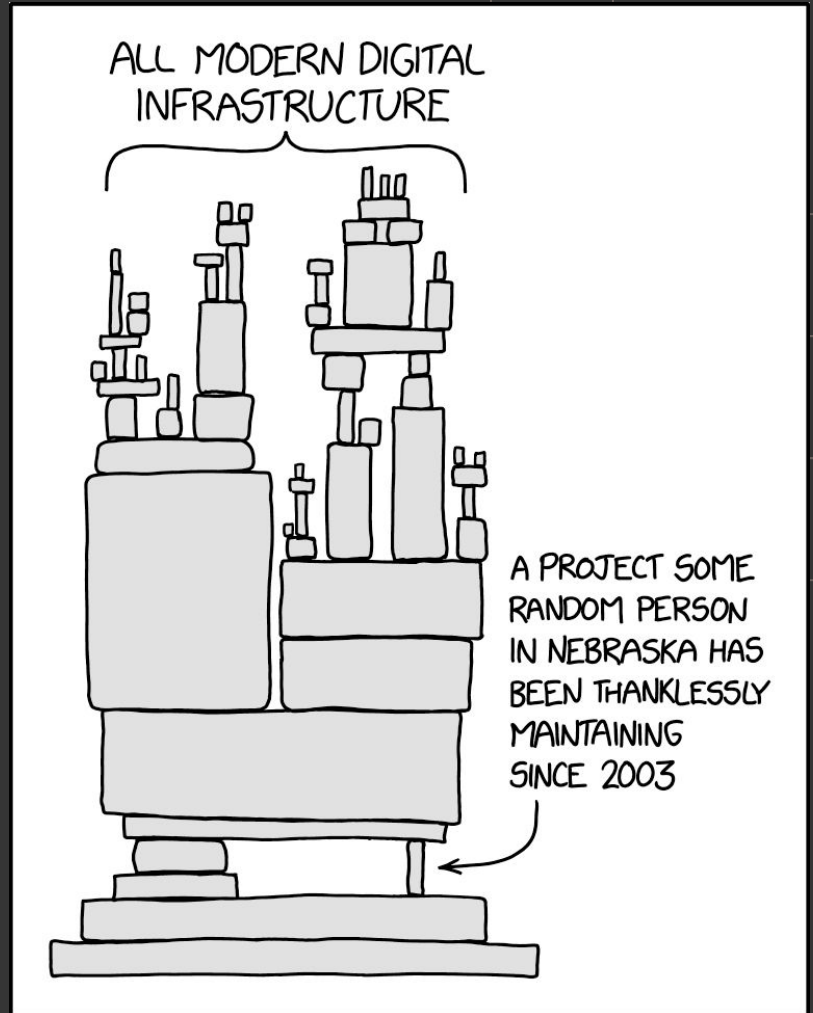


5- Make money

for sustainable OSS



Open source != free





Region: **Germany**



Internet

Recognition of Work on Open-Source as Volunteering in Germany



Henning Lammert



Petition is addressed to



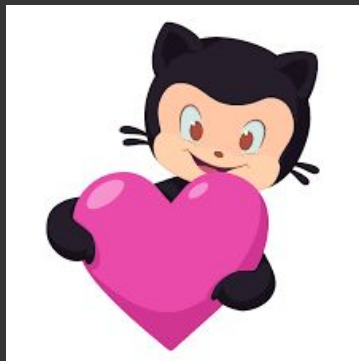
German Bundestag, Petition Committee

8,909 Signatures

8,404 from 30,000 for quorum in Germany

OSS Sponsorship

DRIPS



thanks.dev



PATREON

 **open collective**

Become a sponsor to Samuel Berthe



Samuel Berthe

samber

📍 Nantes, France

Hey! I'm Samuel

I love open source and try to give back to the community.

I create utilities that empower developers worldwide, making Go projects faster to build, easier to maintain, and more enjoyable to work with. Most of my projects originate from the needs of my own work, and I open-source them so that others can benefit as well. As a result, the majority of these projects are battle-tested and run in production.

I currently maintain:

Go toolchain:

- [samber/lo](#): A Lodash-style Go library based on Go 1.18+ Generics (map, filter, contains, find...)
- [samber/do](#): A dependency injection toolkit based on Go 1.18+ Generics
- [samber/mo](#): Monads based on Go 1.18+ Generics (Option, Result, Either...)
- [samber/oops](#): Error handling library with context, assertion, stack trace and source fragments
- [samber/hot](#): In-memory caching library

Select a tier

Monthly

One-time

\$ 20

a month

You'll receive any rewards listed in the \$20 monthly tier. Additionally, a Public Sponsor achievement will be added to your profile.

\$5 a month

Select



\$20 a month

Select



\$50 a month

Select



\$100 a month

Select



GitHub sponsors

Sponsor this project



samber Samuel Berthe



Sponsor

[Learn more about GitHub Sponsors](#)

Current sponsors 8



Past sponsors 15



[lo](#) / [.github](#) / **FUNDING.yml**



samber Create .github/FUNDING.yml

Code

Blame

1 lines (1 loc) · 17 Bytes

```
1  github: [samber]
```

Github Sponsorship



Samuel Berthe <contact@samuel-berthe.fr>

to [redacted] ▾

Hey [redacted]

I just read your article on [redacted] and [Go logging](#). Great write-up! Thanks for referencing my libraries, slog-multi and slog-sampling.

I wanted to see if you'd be interested in a GitHub sponsorship to help promote [redacted] within the Go community?

For [redacted] \$/mo I can feature the [redacted] logo on all my "slog" [libraries](#), with [redacted] visitors per month, and drive meaningful traffic to your app.

[redacted]

Let me know if you'd be interested or if you have any questions.

Kind regards

Samuel

What about accounting ?



Donor

✓ Added to GitHub Invoice

For maintainer

✓ No fees (individuals)



No Invoice



No VAT



No clear status



Auto-entrepreneur

After tax: -50%

Can you live comfortably on GitHub
Sponsors income alone?

Good luck. 🙄



Otherwise, stop asking for funding
and start a business.

Monetize with dual-licensing

An open-source license (Apache 2.0 or MIT)

A commercial license (custom)

Lifetime license or monthly/annual license

Dual-licensing: handle contributions

CLA (Contributor License Agreement): Contributors let you use their code.

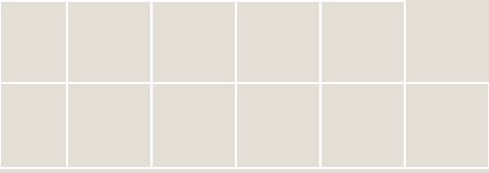
<https://cla-assistant.io>

Monetize with dual-licensing

Use Stripe Billing + Stripe Payment Link + n8n
+ send key by email with unsubscribe link

Sign the activation key with
private key + expiration date

Works offline and in air gap environment



6- Security and hygiene



Supply chain attack



Limit dependencies

Install

```
> npm i is-string
```



Repository

 github.com/inspect-js/is-string

Homepage

 github.com/inspect-js/is-string#readme

♥ Fund this package

⬇ Weekly Downloads

45,192,898



Secure your GitHub account

- MFA
- Disconnect old email addresses
- Disconnect third-party apps
- Restrict token permissions
- Restrict GitHub actions
- SSH keys with passphrase

Overview

Security overview

- Reporting
- Policy
- Advisories

- Vulnerability alerts
- Dependabot
- Code scanning
- Secret scanning

Security policy • Disabled
Define how users should report security vulnerabilities for this repository
[Set up a security policy](#)

Security advisories • Enabled
View or disclose security advisories for this repository
[View security advisories](#)

Private vulnerability reporting • Enabled
Allow users to privately report potential security vulnerabilities
[See reported vulnerabilities](#)

Dependabot alerts • Enabled
Get notified when one of your dependencies has a vulnerability
[View Dependabot alerts](#)

Code scanning alerts • Needs setup
Automatically detect common vulnerability and coding errors
[Set up code scanning](#)

Secret scanning alerts • Enabled

Update dependencies



Renovate



Dependabot

After you pass away

es

Preview

Start export

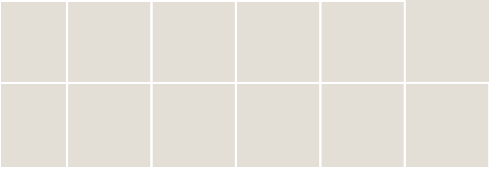
Successor settings

Learn more about account successors.



Revoke

Delete account



7- Last but not least

Conclusion



Key takeaway

Developing an OSS project is not
weekend side-project

But a 10y journey.

Key takeaway

Make it work. Make it right. Make it fast.
(In that order!)

It's ok to start with a bad code quality.
No one will notice.



Thank you !



github.com/samber/talks



github.com/sponsors/samber

SHIFT TIME TO CUSTOM

27 - 29 Mars
2026



“Comment Google Antigravity Transforme le vibe-coding”



Guillaume Vernade

Developer Advocate @ Google Gemini & Deepmind



11 Février 2026 à 19h00



4 Rue Voltaire, 44000 Nantes

chez **_icilundi**